

NORTH CAROLINA MEDICAL JOURNAL.

Number 4. Wilmington, April, 1879. Vol. 3.

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NORTH CAROLINA MEDICAL JOURNAL.

M. J. DeROSSET, M. D.,
THOMAS F. WOOD, M. D., Editors.

Number 4.

Wilmington, April, 1879.

Vol. 3.

ORIGINAL COMMUNICATIONS.

SOMETHING ABOUT DRINKING WATERS.

By ALBERT R. LEDOUX, Ph. D., Chapel Hill, N. C.

Every man must have his vocation; every one his specialty. We are all dependent on others, whenever problems outside of our own line of business or research demand a solution.

Recognizing these truths, every man should feel that he owes a duty to his fellows, and that his motto should not only be “live and let live,” but also “live and help live.”

No science has done more *gratuitously*, for the advancement of the human race, than medicine.

No other vocation *gives away* so much of invention, research, time, labor, money, to make men stronger, happier, better.

Following out her lofty aims, medicine has called to her aid sister sciences, and united with them to build up new safeguards around humanity.

Thus, for example, medicine has united with chemistry and architecture to form “Sanitary Science,” with all its details of work and endeavor for the health of nations, towns, villages and homes.

No question which sanitary science discusses and investigates, is more important than the relation of drinking water to health.

The one grand cry of humanity—yes, of the brute creation, and of the vegetable world too—“is give me something to eat and drink.” Dame Nature furnishes about two hundred and fifty articles to man for food, giving him the greatest variety, from which to choose, when hungry; but, when he would slake his material thirst, she offers simply water. It is the most abundant thing upon earth, as every school-boy knows.

Over two-thirds of our globe is covered with this wonderful liquid—while, on the solid ground, there are comparatively few localities, where water will not be struck on digging. In fact, our soil is one vast sponge, holding in its porous mass—water.

The air around holds water in suspension; the trees and lesser plants hold water in every leaf and branch—while fruits are mainly—water.

Seventy-five of every one hundred pounds of potatoes are—water; one acre of potatoes requires, at the very lowest estimate, twenty tons of water, during the growing season, to bring tops and roots to a perfect healthy maturity.^[1]

Eighty per cent. of apples, pears, peaches, &c., is—water.

Eighty-six per cent. of milk is—water. We ourselves are, by weight, mainly—water.

A body weighing one hundred and twenty pounds, if dried till free from all its liquids, would weigh but twenty pounds—while three-fourths, by weight, of the human body is water. If we were to make a box 16 inches square and the same deep, (a cube of 16 inches,) with walls one inch thick, and fill it with water, the ratio of water to wood would very nearly represent the relative proportions of solid and liquid constituents of the human body, both by weight and volume.

Having taking a hasty glance at the magnitude of the demand of organized nature form water, we will pursue it no farther in a general way, but confine ourselves to *man's* particular need and the character and sources of supply.

To obtain the quantity, which he requires to meet the demands of his system, a man consumes every year, about three-fourths of a ton, or fifteen hundred pounds of water. Some of this supply comes, of course, from waste and vegetables, which are, on an average, three-fourths water, and from bread, which will average 45 per cent. A certain quantity is also generated in the combustion of food, but the greater proportion is taken in drink.

It will suffice to mention one or two of the uses of this water, which plays such an important part in our system. It gives a medium of circulation—of transportation—to solid, inert substances.

As the great oceans and mighty rivers of earth bear upon their bosoms noble ships, freighted with the wealth of nations, so in the blood the precious corpuscles are coursing, borne on in their life-giving, life-sustaining mission, by the water in which they float. So in the milk and other animal creation, water bears safely a freight of valuable, solid particles, or carries off useless solids in solution. It gives pliancy to muscle and flesh, and serves many another purpose in the economy of the human system.

It can be seen from the outlines we have just given, without the need of a further demonstration, that the *quality* of drinking-water is of the utmost importance.

It is strange, but true, that man needs to be protected, even by force, against himself, and this is well exemplified in the matter of drinking-water.

One of the first and last labors of every State or city Board of Health is to prevent men from poisoning their drinking-water or allowing others to do it for them, and to keep them from using it, when it is thus poisoned.

Those who live in the country are often prone to thank God that they live beyond the reach of sewer gases and other poisonous contaminations of city wells; but, before being too confident, let us ask ourselves the question, “are the people of our villages, or on our farms and plantations, entirely free from typhoid fever, diphtheria and other diseases, whose origin is so *often* traced to impure drinking-water.”

But what is pure water? It may surprise some of our readers when we state that *absolutely pure* water, used constantly, is unhealthy!

Distilled water, taken copiously, will soon make one sick. A number of diseases which prevail in some mountain countries are ascribed by many to drinking comparatively pure snow water. Another surprise for some, perhaps! We consider water tasteless, but *had* it no taste we would loathe it! But we will return to this hereafter.

There are two tasteless, odorless, colorless gases called respectively, “Hydrogen” and “Oxygen.” If we mix them in a vessel and apply a match, there is instantly a powerful explosion, heat is generated and there is formed—water. The two strange, invisible gases have combined and formed the well-known liquid. Whenever a substance which contains hydrogen, like wood, paper, starch, sugar, &c., burns, it forms water with oxygen of the air. In fact, these two gases are always ready, on the slightest provocation to unite. Rain, or melted snow, approaches nearest to chemically pure water, but all wells, springs, rivers and seas contain dissolved substances, in greater or less degree. Water being the most universal solvent, whenever it comes in contact with the earth, dissolves the soluble constituents of the soil through which it flows, and hence it is that on analyzing water we find solid substances in solution. Streams in their onward course take up more and more matter; rivers flow into the ocean, and in the ocean the maximum amount of solid matter in solution is found. There the rivers have been

carrying their load for centuries and leaving it, since water evaporating carries nothing away with it. Besides these solid bodies some gases are dissolved by water. Many waters contain, besides mineral and gaseous bodies, organic matter—living animal and vegetable organisms or decayed substances.

Having noticed the different classes of foreign ingredients in water, let us study them a little more closely, considering their effects and influences on the human system.

Mineral Ingredients.—These are the substances most frequently met with in waters, indeed, organic matter may be said to be rare in comparison to the wide distribution of the inorganic solids. An enumeration will reveal many things with which we are more or less familiar. The most common solids in well, river and sea waters, are lime, magnesia, soda, potash, iron, chlorine, sulphurous acid, silica, phosphoric acid and alumina.

An analysis of an average soil will reveal the presence of all of these substances, so that it is very easy to understand how they got into the water. If we arrange some of them in another way, grouping some together, we will recognize many things common in medicine or every day life. Thus we have

Chloride of sodium or common salt.
Sulphate of soda or Glauber's salt.
Sulphate of magnesia or epsom salt.
Sulphate of lime or plaster.
Carbonate of lime or limestone.
Carbonate of soda or common "soda."

The effect upon the system of each one of these substances, or combination of them, when occurring in water is the same as when given in ordinary prescriptions, but there is a point beyond which chemistry has not penetrated; for instance, a glass of some mineral waters, containing but a few grains of solid matter in solution will often produce a quicker and more powerful effect when taken than twice the amount of the solid constituents, shown to be present by analysis, prepared artificially by the apothecary.

The reason of the efficacy of some mineral waters may be in unrecognized combustion of known elements or the presence of substances as yet beyond the power of chemistry to detect.

Four of these common substances mentioned above have a characteristic taste, and as they are all found in nearly every well, it follows that these waters must have *some* taste. This, as already stated, is a fact. We, who are accustomed by constant use to one particular well, fail to recognize any taste at all, while a stranger will often detect it at once. Distilled pure water tastes “flat” and very disagreeable to us, because we miss the salts and gases, which distillation has removed.

A chemist will often work in an atmosphere filled with noxious and powerfully smelling gases, but will not be able to perceive them, though a stranger would not only notice them at once, but with great difficulty endure them.

The mineral constituents in well or river water will average five to thirty grains per gallon; while they vary in amount, as shown from analysis from one-twentieth to twenty thousand grains per gallon.^[2]

Waters containing much lime or magnesia, are called “hard,” and are the only kind found in some sections of our country.

Besides the common mineral constituents of water, there are others, which have been occasionally detected, such as arsenic, barium, strontium, lithium, bromine, iodine, fluorine, zinc, copper, lead, silver, antimony, nickel, cobalt, &c., &c. It will be noticed that many of these are poisonous, but nature suffers their presence, in minute quantities only,—except in rare instances,—while the localities having metallic elements in their waters are few and chiefly among mines and ore-beds. These substances rarely occur in our wells and springs and hence they need no particular consideration here, though, if present, only analysis is a safe means of detection.

Where water contains a large amount of mineral matter, it has a decided effect upon the digestive organs, and, after shorter or longer use, tends to produce diseases, such as dyspepsia, constipation, gravel, &c., &c. But waters with a large enough amount of mineral matter to give them a decided taste, are called “mineral waters” and are used rather as medicine, than as a habitual means of slaking thirst.

The Common Gases dissolved or imprisoned by well and spring water are carbonic acid, sulphuretted hydrogen and the components of air—oxygen and nitrogen. Of these only one is dangerous—sulphuretted hydrogen—and that is easily detected by its smell, resembling that of spoiled eggs. It gives the characteristic odor to “sulphur” waters. Carbonic acid and air give to

waters their sparkling quality—the former being often artificially introduced, as in “soda-water,” &c.

The Organic Matter in well, spring, and river water may be dead or living, and we will consider the two classes separately.

Lifeless Organic Substances.—These may be the remains of organized bodies and plants once living in the water, or animal and vegetable matter from some outside source. The latter is the most frequent source of organic impurity.

In localities where heat and marshes abound, water may often be colored by organic acids and other substances which dissolve, and yet not be appreciably unhealthy. Such waters are common in some countries, and present dangerous *possibilities*, should fermentation and putrefaction at any time set in. Decaying fish and animals, leaves, &c., &c., form the ordinary and accidental organic impurities of water. These are unhealthy not only in themselves, but especially because they offer to the germs of disease or pestilence a harbor and sustenance.

As was stated in the early part of the paper, men must be protected from themselves and especially against water of their own poisoning. Nature's strivings are constantly to make clean the unclean; to dissipate the noxious and to destroy the hurtful, but man by breaking nature's laws, brings ten fold vengeance on his head. The most dangerous poisons in well water are the drainings of sewers, sinks, yards and privies, and the refuse from towns.

These organic, poisonous matters ooze through the soil into wells and springs, and as before said, *may* not show any bad effect for sometime, but sooner or later disease and death will surely visit the unsuspecting household and the physician's aid be sought in vain, for with every draught of water which passes the fevered lips, the sufferer imbibes new poison and hastens the inevitable end. Moreover, the germs of many contagious diseases, which feed on filth and multiply in foul water, are nurtured and preserved in warm climates through winter weather, by the equable temperature of wells and cisterns, and are ready to start anew on their errand of death, when a favorable moment arrives.

The city of Wilmington is no doubt above the general average of Southern cities in sanitary condition, but what a picture the February number of the JOURNAL showed us. Think of it!

“There was one well two feet from the privy, two wells four feet from the privy, thirty-three wells ten feet from the privy, two hundred and twenty wells from twenty to thirty feet!”

The soil upon which Wilmington is located being “nearly as white as the seashore *and as permeable!!*”

It is not our purpose at present to depict the danger of such neglect of sanitary precautions, so much as to point to a remedy.

1st. We say unhesitatingly, if a well shows signs of contamination by sewerage or other like matter, fill it up!

2d. Build all sinks and privies as far as possible from the well.

Through permeable soils and strata, dangerous liquids may ooze to a distance of many feet. We know of cases where wells had been used for years with no bad effect, when suddenly disease and death appeared. The poison, though slow in its course, had finally reached the well and a chemical analysis revealed contamination from privies thirty feet or more distant.

The living organisms which are found in water are, some of them injurious; some beneficial.

Under favorable conditions of light, warmth, &c., countless millions of living things will spring into life in any water; the more impure, the more abundant they will be. If the water is alkaline they will be animalculæ or infusoria: if acid, fungi, algæ, &c.

They are never found in fresh rain water, but abundant in nearly every cistern. The office of infusoria is in water, that of the buzzard on land: they are scavengers, and purify the liquid by feeding upon the decaying matters it contains. But the microscope reveals to us in water, contaminated with sewerage, for instance, minute germs capable of motion, which, as in the case of the infusoria, live on the organic matter, but are believed to accompany if not to cause many forms of contagious disease, filling even the air, in times of epidemic.

To detect many of these impurities and dangers, chemical analysis and the microscope are sometimes indispensable, but the following rules may awaken suspicion and lead to a scientific investigation of the quality of drinking water in some cases.

A good drinking water is perfectly colorless and transparent, without smell or noticeable taste and agreeable to the palate. It should not lose its clearness in boiling and should leave a very small residue on evaporation.

Where impurities are suspected, an analysis should be obtained if possible, if not, filtering through charcoal or sand, or boiling will often either remove or render harmless various dangerous ingredients.

Our State Board of Health have done the people of Wilmington and of the whole State a great service in directing attention to this subject. May they go on and prove a mighty blessing to the Old North State. Let us give them the aid and encouragement they deserve.

SUBSTITUTE FOR COD-LIVER OIL.

An excellent substitute [for cod-liver oil] and one often better tolerated, is the fat of pork properly prepared. I direct a thick portion of the rib piece, free from lean, to be selected and allowed to remain in soak for thirty-six hours before being boiled, the water being frequently changed to get rid of the salt. It should be boiled slowly, and thoroughly cooked, and while boiling, the water must be changed several times by pouring it off, and fresh water nearly boiling substituted. It is to be eaten cold in the form of a sandwich made from stale bread, and both should be cut as thin as possible. It is very nutritious, but it should only be given in small quantities until a taste for it is acquired. It is the most concentrated form in which food can be taken in the same bulk, and I have frequently seen it retained when the stomach was so irritable that other substances would be rejected. For this condition of the stomach it may be rubbed up thoroughly in a porcelain mortar and then given in minute quantities at a time. It is made more palatable by the addition of a little table salt, and this will be tolerated, while the salt used for preserving the meat having become rancid, if not soaked out, will produce disturbance even in a healthy stomach. I, some years ago, saved the lives of two of my children, who, on different occasions were suffering from cholera infantum, by feeding them entirely on the fat of pork prepared in the way I have described, and, while nothing else would be retained in their stomachs, not only was it retained, but it also had a beneficial effect on the diarrhœa.—*Emmett's Prin. and Prac. of Gynæcology*, p. 102.

A surgeon in London was recently tried and convicted of manslaughter, for not heeding numerous calls from a patient who afterwards died apparently from this neglect.

COUNTRY CLINQUES.

VI—A CASE OF OPIUM POISONING. (?)

BY A NORTH CAROLINA PHYSICIAN.

Katie L., colored, æt. 40, an expert and industrious laundress, but a woman of lewd character, has been under observation for several years. A reliable history of her previous life I cannot give. According to her own account, she had suffered almost every ill to which flesh is heir, excepting gonorrhœa and syphilis. There was a marked systolic murmur over the base of the heart, which, since she showed no other symptom of anæmia, I considered indicative of structural lesion; but as will be seen, I was probably mistaken in this opinion.

The most interesting feature in her case, and for this I most often prescribed, was the concurrence of epileptiform convulsions with every menstrual epoch. For six years has this occurred with almost uniform regularity, an occasional intermission only, having been brought about as the result of medical treatment.

En passant, a word may here be said against the too generally accepted idea that albumen found in the urine of puerperal women, after convulsions, is an indication of a previously existing albuminuria. On five successive occasions, I examined the urine passed by this woman before the occurrence of convulsions, and within a few hours of the attack.

There was not a trace of Albumen. Invariably I found the urine which was passed *after* the epileptic seizure to be *highly albuminous*. It gradually resumed its normal character in from two to six days, in a direct ratio to the severity of the attack. Again, the severity of the convulsions maintained an inverse proportion to the quantity of the menstrual discharge. When this was profuse the attack was light, when scanty, more severe. The convulsions generally appeared just before, or at the beginning of the monthly flow. Latterly their occurrence has been somewhat irregular, as has also been the case with the menses. Elaterium in $\frac{1}{4}$ grain doses, frequently cut them short, but exhausted the patient to such an extent that it had to be discontinued. For several months past I have been controlling the convulsions with $\frac{1}{2}$ grain doses of morphia per orem, repeating every hour

until relieved. She has frequently taken two, and a few weeks ago took three such doses, without exhibiting symptoms of marked narcosis.

At 9 A. M., on February 25th, I was called to see her. She had had four most violent convulsions during the previous night, and was complaining of terrible pain in the head, with nausea and vomiting. She expressed the conviction that another convulsion was imminent, and begged for relief. I immediately and without hesitation introduced $\frac{1}{2}$ grain of hydrochlorate of morphia under the skin of the forearm, and having other engagements, left her. At 12 M., I was sent for, and informed that shortly after my departure, she sank into a deep sleep with stertorous breathing. All efforts to rouse her, had failed. On examination, she presented the following symptoms: There was total insensibility, except a slight twitching of the eye-lids when the conjunctiva was touched. The pupils were contracted to the size of a pin's head. Respiration was shallow, irregular and interrupted, and numbered ten to twelve per minute. The extremities were cool and the face somewhat cyanosed. The pulse beat regularly, though feebly, 110 per minute. To my surprise, auscultation showed the *absence* of all adventitious sounds over the region of the heart.

Despite the gravity of the symptoms, I felt only a slight degree of alarm, when I considered the improbability of so small a dose of morphia proving fatal. Being compelled to leave, I merely directed the attendants to keep up circulation, by friction of the extremities. At 3 P. M., the condition of patient was unchanged, except that the extremities were more difficult to keep warm. Temperature in the axilla was 97.4° . The breathing was not at all better, and insensibility was, if possible, even more profound than at my previous visit. I injected 1–20th grain sulph. of atropia under the skin of the forearm, and during the next hour I made frequent applications of a moderately strong galvano-faradic current, one pole being placed in the epigastrium and moved along the insertion of the diaphragm, while the other was pressed upon the middle of the neck just behind the sternomastoid muscle. The heart's beat was temporarily strengthened, and respiration slightly increased in depth and frequency by each application. At 4 P. M., I injected 1–12th grain of atropia, continuing the use of electricity. At 5 P. M., the circulation appeared to be failing, the pulse being decidedly weaker and the extremities cold. Respiration was about 15 per minute,

irregular and shallow. The pupils were still obstinately contracted. I now injected 1–6th grain of atropia and placed a bottle of hot water under each arm, and a large jug to the feet, still employing electricity at intervals. At 6 P. M., the change in my patient was evidently for the worse. To be sure the body was warm (100° F.), but the pulse at the wrist could only irregularly be felt. The heart contracted feebly but regularly 115 times per minute. Respiration was more shallow, although now 18 to 20 per minute. The pupils were unchanged, and there was absolute insensibility of the conjunctiva. I now injected $\frac{1}{4}$ grain of atropia. In twenty minutes the effect of this dose was perceptible. The pupils were widely dilated, and respiration increased to 30 per minute; but alas, the heart, although it contracted 130 to 140 times per minute, failed to convey even the slightest impulse to the wrist. Cyanosis had disappeared but the insensibility continued.

During the next seven hours, I injected into the bowel $\frac{1}{2}$ oz. of whiskey every half hour. All of it was retained. During this time, the patient occasionally made an unconscious effort to swallow the mucus which accumulated in the fauces, and succeeded so far as to diminish temporarily the rattling and gurgling which now accompanied every respiration. Several times after this effort at swallowing, respiration had to be stimulated by the electrical current. At midnight there was a slight convulsion, after recovery from which the patient again lapsed into the same condition. Gradually there was an increase in the rate both of circulation and respiration, until at 4 A. M., the heart beat 150, and the breathing was 36 per minute. There was no dicotism, but the heart's contraction was steadily becoming more feeble and imperfect. The pupils were still widely dilated, the extremities warm, and the temperature 101°. Fifteen minutes later the heart ceased to beat, and death supervened without a struggle.

For my own sake, as well as for the good of the profession, I invite the most rigid criticism of the above report. The case in many ways is both interesting and instructive. Was this a case of opium poisoning? The symptoms appear to answer this question in the affirmative. I have so frequently given a similar, and even a larger dose in pressing emergencies, without the least unpleasant effect, that I find it difficult to realize the fact that this patient was fatally poisoned by $\frac{1}{2}$ grain of morphia. Such an unlooked for result has given a terrible shock to my confidence in the safety of large doses of morphia under any circumstances. The heart lesion of it before existed, evidently did not influence the result, as all signs of it were

gone when I examined, three hours after the administration of the morphia, and they were not reproduced, even under the stimulation of electricity, atropia and whiskey.

Did I give too little atropia? Three doses of 1–12th grain each, sufficed to counteract the poisonous influence of 1½ ozs. tinct. opii, in a case which presented much graver symptoms of poisoning, (vide pp. 65 and 66 N. C. MED. JOUR., Feb., 1879). Was I too slow in administering the antidote? In the present case the use of atropia was commenced six hours after the morphia was exhibited,—in the case above cited five hours elapsed before any atropia was given. In the case I previously reported, an aggregate of ¼ grain was given within seven hours after the opiate was taken, in the present case ½ grain within eight hours. Did I give too much atropia? At a single dose, Dr. Fothergill gave 1 grain in a similar case, and the patient recovered. ^[3] (*Antagonism of Medicines*, p. 133). Should I have given digitalis or strychnia hypodermically to further stimulate the heart? That poor organ appeared to be doing its best, and to tell the truth, I felt that I had had enough of hypodermic medication for one day, and felt unwilling to risk anything more, after being so disappointed in my expectation of relief from atropia. I am open to conviction upon any one or all of the questions I have propounded.

In the light of our present knowledge of laceration of the cervix uteri, Dr. Whitehead's article on "Hypertrophic Elongation of the Cervix Uteri" (Trans. N. C. Medical Society, 1875, p. 90), has peculiar significance.

LARYNGO-TRACHEOTOMY.

By CHARLES DUFFY, SR., M. D., Catherine Lake, N. C.

Read before the Onslow County Medical Society, September, 1878.

Gentlemen of the Onslow County Medical Society:—

I was written to sometime ago, by a member of the State Medical Society, asking my views in regard to operating on the windpipe. My experience in such operation has been very limited, six times being the maximum of my labors in that direction. My first was a failure, done for the relief of cynanche trachealis, the operation being performed too late.

The other five cases succeeded admirably; four of the patients ranging from eight months to three years old, the other a woman of middle age. The first of these cases was operated on for the removal of a watermelon seed. The child was less than 2½ years old, and was very fat, so much so that the depth from the surface, would seem to forbid approach to the external surface of the trachea, still less to the internal, but by patience and perseverance these difficulties were both overcome, and respiration rendered comparatively easy. The next idea, was to get out the seed, and one attempt after another was made to no purpose, the wound inclined to close at the same time. I next lengthened the incision, and the sides of the wound were well drawn apart. My next step was to trim off the sides or edges of the cartilages; this being done, gave the seed a fine opportunity to present itself, and the child was placed in a cradle and diligently watched, with orders to take him in the arms and walk about with him, in case of difficulty of breathing coming on, which had to be done from time to time. The seed was expelled through the aperture to our great joy and gratification, several hours after the last step of the operation. The child was a son of Mr. Thomas Holland, of this county. He grew to adult age, and was killed by a horse running away.

From this case I learned that the removal of foreign substances by forceps or other instruments, except they are metallic substances is seldom necessary, there would be much more difficulty in retaining them or

preventing their escape. As soon as the windpipe is cut into there is a rush of wind that follows, that moves the substance by the double ability or means of respiration, caused or provided by the operation, and the next we know the substance is expelled. Certain it is, it is not going to stay there, if there is room for its escape and the patient is rightly attended to. When certain that all has come away, apply adhesive plaster drawing the parts together, a stitch or two might be necessary in some cases, it soon gets well.

My next case was the woman alluded to, the wife of Mr. Amos Wooten, of New Hanover county. A piece of beef gristle got into the wrong passage. After several spasms, and vain attempts to get it out she sent for me. I got to her as soon as possible—the distance being sixteen or seventeen miles. On enquiry I learned the particulars of her case. I found her composed. I told her it might not be in the windpipe, and we had better be certain about it. I passed a probang down the œsophagus and found that it was not there. After waiting a little longer, she had a violent spasm that hurried and increased her determination to have it out. So violent was the spasm, that it created doubts on her mind as to her chances of living, or of being able to bear up under the operation. She next turned her head toward me and remarked that she was ready. I had no medical assistant with me. I operated without chloroform—the woman fainted. There was camphorated spirits close by, and I sprinkled it heavily and forcibly in her face and over her chest, and rubbed some in her mouth. She revives with a vim and sends the gristle forcibly, not only out of her mouth, but nearly out of doors, rejoicing all hands around.

I applied sticking plaster and left; saw her in a few days; she was well.

My next operation was on the child of Mr. Enoch Foy, who had the misfortune to get a watermelon seed in his windpipe. The usual symptoms occurring, he came on with his little boy and had him relieved—the seed coming out several hours after the operation.

The next was a child of Mr. Marshall, (another fine boy), another case of watermelon seed, which was operated on with like success.

My last case was a child 8 or 9 months old, a very pretty and fine little girl, the daughter of a Mr. Padjet of this county. She had been playing with an ear of corn, given to amuse her; some of the grains coming off and one and a half getting into the windpipe, as shown by the sequel. She was operated on, assisted by Drs. Cox and Nicholson. The foreign substance did

not come out as soon after the operation as the other cases. The wound was not kept open by the attendants, and in consequence I had to re-visit, reöpen and somewhat enlarge the incision which was attended with the usual good results. The child was very fat, and the space for operating in so young a child, was necessarily very limited. One grain of corn and the part of another was expelled. I will next give my “modus operandi,” or rather my imperfect manner of operating.

The patient being laid on a suitable table, with the chest elevated, by placing a pillow or folds of cloth underneath. The head is next laid back neatly observing the direction of the mesial line strictly, and throughout the operation. The instruments previously got ready, and those which I prefer, are a scalpel with a sharp handle, a director and probe, two bistouries, one sharp and the other button pointed, a forceps, tenacula, sponge and ligatures. But so far I have never needed the ligatures. I have always stopped any little bleeding that occurred by applying a pencil of nitrate of silver. All these ready, also a basin of cold water, standing on the right of my patient, I place the finger and thumb of my left hand, one on each side of the thyroid cartilage, and commence my first incision from its lower third if a child, and from its lower edge if an adult, for obvious reasons, namely: In the child we want room, and if necessary can enlarge the incision in that direction, with but little difficulty, the cartilage affording no resistance. In the adult we have more room, and the cartilage is often found hard, and unyielding in persons of advanced life, and it is therefore necessary when enlargement is required in the adult, to cut an additional ring or more of the trachea. I continue my incision below the cricoid cartilage, so far as one or more of the rings of the trachea. The track of the operation being now laid off, I proceed cautiously, an assistant sponging, and applying caustic, as may be necessary to arrest any little bleeding that may ensue, whilst I, with the handle of my knife, push aside any vessel likely to bleed—cricoid artery or otherwise. I next lay hold on the cellular sheath of the trachea, at the lower edge of the track of my operation, and at this point I enter with a sharp pointed bistoury, holding it close to the point, and cutting upward not more than one-eighth of an inch and withdraw it in favor of the button pointed bistoury, with which I slit upward the windpipe, as far as the starting point of the first incision—not moving the instrument back and forth, but holding it perfectly steady, carrying it or rather pushing it, aided by the other hand from below upward, with the handle of the knife inclined

downward. The operation now done, is made known by a whizzing which it is necessary to look after, and as all-important. I consider it the safety valve of the patient.

This operation may also be performed from above, downwards, with a sharp pointed bistoury, holding it not far from the point; the forefinger on the back of the knife—taking care to help the cricoid artery out of the way, which I have always been able to control when cut, by the application of nitrate of silver. The patient may be, if necessary, turned on the side to prevent blood from passing into the windpipe.

I begin close by the lower edge of the thyroid cartilage, and carry it so far as the second ring of the trachea; but in either case, whether I open upward or downward, the tenaculum can materially assist in the operation, by drawing down the tube when cutting upward, or by drawing upward when cutting downward—the hook to enter behind the knife in either case.

The use of the hook is most necessary when operating on young children. The object in pushing the knife, holding it steadily, is from knowing that it long since has been found, that an artery will give way before a knife when carried in this way that might otherwise have been cut immediately by a “see-saw” motion.

After the operation is performed, I direct the attendants to keep the opening clear of obstruction—bloody froth, &c., or anything that may make its appearance in the wound. Artificial respiration must be kept up until the foreign substance is expelled or removed. A probe or knitting needle will suffice for that purpose, one or the other must be used several times a day and night, in fact as often as needed; I use no gauze, it might get sucked to, or drawn into the opening, and thereby defeat the intent of the operation. In cases needing the use of the canula I make no reference.

I prefer laryngo-tracheotomy, sometimes denominated circo-tracheotomy, which I have been endeavoring to describe, to any other, for all ordinary purposes. We have less risk, and more room, and it is more adapted to the relief of children and might with propriety be called the higher operation to distinguish it from tracheotomy, which rightly speaking is the lower operation. This would draw a distinction between the two, and it is necessary that line should be observed, and that when these operations are spoken of, we should know what importance to attach in either case, and give to either operation the degree of approbation it may deserve.

I cannot close this subject without giving the opinion of a very able anatomist regarding it, Harrison, of Dublin. In the first place he speaks of an irregular artery, which he has seen running along the front of the trachea to the thyroid gland and cellular membranes beneath it. He had seen this so frequently in this situation, that he describes it under the name of the middle thyroid artery. "This is" he says "so common an occurrence that it should be remembered by the practitioner of tracheotomy." He further goes on to say, "in children the space for tracheotomy is very limited," and directly that "particular attention be paid to the inconsiderable portion of the trachea that can be exposed between the thyroid gland above, the arteria innominata, the left carotid artery, the remainder of the thymus gland below. The deep thyroid veins also descending to the vena innominata obscure the trachea very much, these together with the great mobility of this tube, add to the danger and difficulty of this operation." Pancoast says: "The checking of hemorrhage from the veins and arteries divided in tracheotomy requires particular attention; from six to eight ligatures are usually employed. They should be applied in general as the vessels are cut and before the opening of the trachea as there must be blood drawn by respiration into the trachea and thereby endanger life."

These dangers constitute shoals and quicksands to the anatomist and surgeon, that has made many a one shudder at their approach. The six or eight vessels to tie, before daring to open the trachea, causes delay dangerous to life, as well as to the success of the operation, and brings into question the propriety of the operation, and sometimes the skill of the physician. In the upper operation, laryngo-tracheotomy, you can enlarge the opening upward whenever necessary, with but little risk, by cutting through the thyroid cartilage. In fact, it may be opened above or below, one or both, with but little risk; whereas in the lower operation it is almost impossible to do so. When it becomes necessary, the safest plan is to enlarge the opening upward, as much as is practicable, and downward as little as we are able to get along with. The space taken up by the lower operation on children is very limited, and the operator must necessarily be cramped for want of room. The cervical portion of the adult trachea is laid down at from two to two and one half inches long. It is composed 18 or 20 fibro cartilages, this makes the space between each ring 1-8th of an inch. According to that measurement, allowing the 20 rings for 2½ inches makes the space taken up by cutting three rings 3-8ths of an inch long in the adult, if no more is

divided, and proportionately less in the child. We can readily understand that those operating in this region do as little cutting as possible, and although the operation so far as the outside incision, may begin at the cricoid cartilage, and terminate as at a little distance from the fossa at the top of the sternum. I have no idea that the trachea is often laid open to that extent. Pancoast directs, "that after separating the two sterno-thyroid muscles, partly with the point and partly with the handle of the knife, and finding no large vessels in the way, pushes up, or if necessary divides the isthmus of the thyroid gland." The next cutting he speaks of, is, "that of the third, fourth and fifth rings, puncturing the tube, with the point of the knife below the fifth ring." He then speaks of running the scalpel upwards with the handle inclined to the sternum, so as to avoid injuring the posterior wall of the trachea. It is easy to perceive in the practice of the present day, that this operation is done for, and best suited to the insertion of the canula, and that the opening of the third, fourth and fifth rings of the trachea can, when divided, answer by binding the canula, a much better purpose than a larger opening, which would allow it to move about, thereby incurring the danger of displacement.

The word tracheotomy as a general term does harm. We ought rather to particularize, and make known on what part of that tube we operate, and not speak of tracheotomy as though it were of little moment in the performance, and that one part of the windpipe cut into, was as much a tracheotomy as another; not by any means should this be thought. I consider that tracheotomy strictly, and according to the definitions of anatomy and surgery, is one of the most dangerous that come within the province of the surgeon; and, on the contrary, I consider laryngo-tracheotomy, or crico-tracheotomy as it is sometimes denominated, a very simple operation, and only requiring ordinary tact in the performance.

Since the above article was written, this operation has been successfully performed by Dr. J. L. Nicholson, assisted by myself and Dr. C. Thompson.

MR. GRANT GIVES THE FOLLOWING RULES ABOUT FOREIGN BODIES IN THE EXTERNAL EAR.

1. Be sure that the foreign body is *seen*. To attempt to extract a foreign body without first seeing it is highly dangerous.

2. Determine what the body is, and, if possible, obtain a sample of the body supposed to be in the ear.

3. Remember that a body which will not swell, and has no cutting edge, will generally remain without causing any urgent symptoms.

4. Seeing the body, determine with a probe if it be movable. If easily movable, concussion with a downward position of ear will often remove it.

5. Warm water injection is the best of all methods of removing foreign bodies.

6. If it be a vegetable substance, do not inject fluid unless you have time to extract the body either at one operation, or shortly afterwards.

7. Injection failing, which is very exceptional, a surgeon, with the necessary appliances, ought to be at once consulted, or should urgent symptoms arise from the irritation in the attempted extraction, the extraction by the incisions, galvano-cautery, boring out by trephine or conical file the centre of substance, and so causing its collapse; or even detachment of the auricle may be necessary.—*The Medical Press and Circular*.

SELECTED PAPERS.

THE YELLOW FEVER AT HAVANA—ITS NATURE AND TREATMENT.

By CHARLES BELOT.

(Concluded from page 165).

It is at the moment even, of this remission, that sulphate of quinine must be administered in a dose of thirty-six grains taken at once in a half cup of black coffee without sugar. When the intermission is complete, its action is marvellous, the disease is immediately moderated; but if there is no remission, it is necessary to be prudent, for sulphate of quinine, because of its powerful action, can do much harm, if it is not indicated. When the disease commences with chills, followed by abundant sweats after the emetic and purgative, there is assurance, that there will be another remission, and then the sulphate of quinine is preëminently the remedy. But when there are no chills in the commencement of the disease, when the prominent symptoms are heat and dryness of the skin, and the fever continues, the exacerbation will not be long delayed, and no propitious moment can be found to administer the anti-periodic.

In cases where sulphate of quinine cannot be employed, calomel is an excellent remedy, especially when in the absence of remission, the tongue shows itself humid, loaded, white, large, the gums engorged, the stools difficult, or when there is bilious diarrhœa. Under these circumstances, calomel taken in purgative doses every half hour, until the characteristic stools of this remedy appear. Very often, after the administration of calomel, remission of the fever and of the congestive symptoms takes place; the skin becomes moist, and sulphate of quinine, the effect of which will be more sure in proportion to the distinctness of the remission, may then be appropriately used. Its effect is assisted by oil and by emollient injections. If there was no chill in the commencement, aconite and tincture of digitalis will be pressed. These are ordinarily sufficient to bring the patient into full convalescence.

The action of calomel and of sulphate of quinine has led some medical men to employ these remedies, united in the same formula. I have never

understood the effect expected from this, and my experience has not proved the result satisfactory. When one is fortunate enough to have determined an intermission, frictions of sulphate of quinine produce a good effect. Calomel is not indicated, when the disease following its course, the alterations of the coat of the stomach become more observable, the epigastric pain more violent. In this case recourse may be had to the treatment heretofore recommended.

We have seen already, that there are three kinds of black vomit—that formed by bile, that formed by decomposed blood, and finally, that which is a mixture of these substances. The most numerous cases of cure are those, in which the black substance is formed by bile, and in these bicarbonate of soda and nux vomica will ordinarily triumph. But if the black vomit is formed from decomposed blood, there is no treatment which can result in positive action.

When cerebral symptoms predominate, with delirium, restlessness, etc., recourse should be had to blisters, to compresses upon the forehead of cold water and of brandy with belladonna, and to the administration internally of calomel alone, or combined with opium. This combination is valuable—it calms the cerebral excitement. When the delirium is violent, it is dissipated or quieted by an infusion of valerian.

Hiccough may be arrested by compressing with the fingers the phrenic nerve on the level of the os hyoid. In other instances, it yields to the application of cold water on the stomach to opium, to belladonna in small doses, to ice swallowed in small pieces.

When hemorrhages occur, the most appropriate remedies are acids, astringents, and iron. Local, such as buccal hemorrhages, yield to lotions of diluted sulphuric acid every half hour, or to gargles of borax. The best means of resisting epistaxis is by application of ice to the forehead; at the same time, by acid injections into the nostrils. Plugging of the anterior openings of the nostrils would be insufficient, because the hemorrhage is passive, and comes from the whole surface of the nasal mucous membrane. For anal hemorrhages, acid injections, the application of cold to the abdomen, and tannin given internally, are prescribed.

General hemorrhage, that is to say decomposition of the blood, is combatted with tannin or perchloride of iron administered every two hours, with lotions of vinegar or of wine of cinchona, applied over the whole body.

Hematuria is combatted with weak sulphuric lemonade.

It is during the period of hemorrhages, that the parotids become swollen—a frequent indication of amelioration of the general condition. When pain or swelling appears, I apply tincture of iodine three times a day externally, *loco dolenti*. During the epidemic of 1862, I had twenty-nine cases of inflammation of one parotid, and seven of both parotids, and lost but one patient. I attribute this success to iodine. When suppuration does not invade the whole gland, premature incisions must be avoided. They would produce the serious inconvenience of retarding the cure, of making the patient suffer uselessly, and of occasioning hemorrhages difficult to arrest.

The tumors which show themselves on different parts of the body ought to be treated with topical tonics. Compresses soaked in wine of cinchona, facilitate resolution. It is not necessary to open these tumors; this would expose them to hemorrhage.

During the first period, slightly acidulated drinks are prescribed, warmed to promote diaphoresis; in the second, the patient takes cool beverages; in the third, tonics are preferable.

During the whole course of the disease, absolute diet is essential. There must be no indulgence on this point. Often a little broth, given before the period of remission, is enough to bring on indigestion, then reëction, and finally death. In the second period, there is sometimes a sensation of false hunger, which deceives the patient; but the least compliance on the part of the physician might be fatal.

To be more exact, we now proceed to examine singly each one of the recognized therapeutic remedies against yellow fever:

Bleeding.—I consider general bleeding injurious, except with individuals of apoplectic temperament, presenting symptoms of inflammatory fever. I repeat that cupping is preferable to leeches. I have already so insisted on this mode of application, that it is useless to allude to it again. I will say only, that it is necessary to employ the spring scarificator, and never the lancet or bistoury.

Pediluvia.—Foot-baths are perfectly associated with cuppings, to diminish local congestions. They ought to be given in the manner following: The patient lying on the back, draws his thighs upon the stomach, the legs upon the thighs. In this position the feet and legs are placed in a vessel filled with warm water, the temperature of which is

gradually increased, until it becomes unendurable. The bath ought to last from fifteen to twenty minutes. Its effect will be increased by the addition of powdered mustard. When taken from the water, the feet should be carefully dried, and mustard plasters applied to the thighs and allowed to remain as long as the patient can bear them, when they will be removed, and placed on the calves of the legs.

Emetics.—Emesis is one of the most important remedies in the first, but becomes injurious in the second period. Black vomit often comes soon after the administration of an ill-judged emetic. The principle indications are these—the tongue humid, saburral, charged with whitish mucous deposits, nausea, disposition to vomit, bad taste in the mouth, the temperament bilious, constitution lymphatic, atmosphere damp, etc. Administered under these circumstances in the first period of the invasion, the emetic is a heroic remedy. I prescribe thirty grains of ipecac, dissolved in six ounces of distilled water, taken in one draught. Nausea soon occurs, and as soon as the patient begins to vomit, the effect of the medicine is assisted by drinks of warm water. The food contents of the stomach are first ejected, then bile. The drinks of warm water should be continued, until the liquid ejected is as clear as the water that is swallowed. After the vomiting, the patient takes one or two cups of tilia. Ordinarily, the congestive symptoms of the brain increase, by reason of these efforts, but after a short repose and a little sleep, the skin becomes covered with sweat, and on awaking, the pain in the head is sensibly diminished.

I prefer ipecac to tartar emetic, because the action of the former is more gentle and more constant, and because tartar emetic irritates the mucous membrane of the stomach. After ipecac, the patient remains calm, whilst after tartar emetic, the nausea continues, and is very often followed by diarrhœa. I insist strongly on patiently awaiting the effect of the ipecac before giving warm water, because prematurely swallowed, this embarrasses instead of promoting the effect of the medicine.

Let us, however, observe, that ipecac, if useful when clearly indicated, may produce deplorable consequences, if administered despite counter-indications. In my experience, it is counter-indicated, whenever the period of invasion is passed, and even during the period of invasion, when the patient has not been attacked immediately after a meal, when the disease has not commenced with chills, when the individual is plethoric and is

subject to cerebral congestions, or when he complains of pains in the stomach, even when fasting.

The first twenty-four hours of the invasion passed, the emetic can have fatal effects. At this period, indeed, the stomach and abdominal organs suffer in a manner more direct, and the efforts of vomiting increasing, these local congestions may determine a condition as much more difficult to encounter, as the period of the disease is advanced. How often have I seen black vomit appear after an emetic improperly given! I have seen under the same circumstances epistaxis which could not be arrested, and such irritability of the stomach, that it could not bear anything.

Purgatives.—Purgatives are of as great importance as emetics in the treatment of yellow fever.

After the administration of the emetic and a repose of twelve to twenty-four hours, the purgative may be used to induce action by the intestines. It slightly excites the secretion of mucus, and facilitates the circulation and the passing of bile by the stools. Drastics should be absolutely excluded, their too violent action producing injurious irritation. Among the purgatives I recommend, above all, castor oil alone, or associated with oil of sweet almonds. When judged proper to be administered alone, two ounces at least should be given at once. When mixed with oil of sweet almonds, three ounces of the first, and two ounces of the second, adding some drops of lemon juice, unless it is preferred to give the patient a slice of lemon, after the potion, to prevent vomiting.

The action of castor oil is a little slow, but it should be assisted with injections of olive oil and warm water.

If the patient has an antipathy to castor oil, sulphate of magnesia is administered, in a dose of one ounce in a half tumbler of fresh water, with the addition of six grains of nitrate of potash. The mixture of these two remedies produces secretion of intestinal mucus, acts upon the kidneys, augments the secretion of urine, and at the same time excites diaphoresis.

This mixture ought to be given in small doses, every half hour, as the stomach will bear it better. Given in this way it is sometimes vomited in part, so that there should be no attempt to give the remainder. If its action is delayed, it should be assisted with injections slightly purgative, warm sea water, or sulphate of magnesia, mixed with olive oil. During the action of

the purgatives, especially of sulphate of magnesia, the patient may drink as much fresh water as he wishes.

Obstinate constipation, indicating a congestive state of the brain, will be combatted with Seidlitz water. The different purgatives generally bring on a calm, and marked relief.

Other purgative substances are employed, among which I would cite rhubarb and Seidlitz powder, or Seidlitz water. They are particularly indicated in the jaundice of the second, and in the commencement of the third period.

The counter-indications of purgatives are colliquative diarrhœa, the third period of the disease, hemorrhages, especially those of the anus. Feeble and lymphatic temperaments do not endure them well. They should be given to women and children with caution, and in small doses frequently repeated.

Calomel.—We have spoken of calomel, the action of which, so different according to the dose, is here formally indicated, either as a purgative, or as an alterative, or as a derivative. We have seen, that emetics were counter-indicated with persons too robust and disposed to cerebral congestion. It is in these cases, that calomel should be employed, at least, when there are no symptoms of local irritation or inflammation of the stomach. But if the tongue is loaded, humid, large and saburral, without redness of its borders, if the region of the liver is painful, the indication for calomel is more precise.

For a purgative dose there will be given every half hour from three to six grains, until the patient has taken eighteen. If, after having obtained the purgative action, it is necessary to continue the medicine for a certain time in smaller doses, as a derivative, it is given in doses of two grains every hour, until the characteristic greenish stools are obtained.

Calomel is still applicable, when constipation persists, despite the employment of sulphate of magnesia. Administered then, every hour, in grain doses up to twelve, it works marvellous effects. I have seen convulsive symptoms disappear after the administration of this remedy. One of its inconveniences is ptyalism, but this is obviated with the aid of Seidlitz powder. Very often when it has been given after the emetic and sulphate of magnesia, calomel is sufficient to produce a remission, when there has been none.

Sudorifics.—When the remedies, of which we have spoken heretofore, do not bring on a remission of the symptoms, it is well to look to sudorifics, which, facilitating the peripheric circulation, produce a general relaxation, and with this an abatement of the pulse. Among the sudorifics, I commonly select Dover's powder, and the liquid acetate of ammonia or spirit of mindererus.

The latter administered in doses of twenty drops, in four ounces of flower of elder, acts as an antiseptic and sudorific. It is indicated, when the skin is dry, with the sharp heat so common in yellow fever of the continued acute type, without remission. I have often seen individuals stubborn to sweat, despite the purgative, transpire abundantly after some doses of this medicine.

Dover's powder suits, when the patient has dry skin, is restless, and turns over in his bed uttering deep sighs. I give for the dose, every two hours, from three to four grains in two or three spoonfuls of infusion of tilia, warm or hot. After the second or third dose, the patient becomes more calm, sleeps, and wakes covered with sweat. The effect of sudorifics will always be assisted by a mustard foot-bath.

Some medical men have considered the transpiration so useful, that they have made it the basis of their treatment. In the outset, they prescribe a steam bath. I have tried this but without advantageous results.

When transpiration is determined, the pulse remains full and strong, diuretics are indicated, and among these powder of digitalis associated with nitre.

Sulphate of Quinine.—Sulphate of quinine is one of the most powerful and most useful remedies in the treatment of this disease; but it must be well indicated, well administered, and in a suitable dose. What are the indications of sulphate of quinine in yellow fever? There must be at least remission, if there is not complete intermission of the fever. Its application is then excluded from the continued type. When the fever has yielded, by the use of the medicines of which we have spoken, or when, with sweat or moisture of the skin, the pulse has sensibly lowered, the employment of quinine is always good. Its effect will be shown, for the strongest reason, in the intermittent type. In this last case, it acts with the same precision and the same success, as in simple intermittent fever.

As sulphate of quinine has a prompt and durable action, the mucous membrane of the stomach ought to be in the best possible condition for absorption. It must, therefore, be empty. An emetic and a purgative, at least the latter, should precede the administration of sulphate of quinine. I know that there are medical men, who administer quinine in the height of the fever, regardless of the state of the stomach and of the mucous membrane. If they have found this treatment beneficial, it is by chance; for it is illogical, and its effects are commonly deplorable.

As for the dose in which sulphate of quinine ought to be prescribed, it depends on the age, sex, and temperament of the patients. For adults and men, the average is twenty grains in a single dose, in about three ounces of black coffee, without sugar. If it is feared, that the irritated stomach cannot bear so strong a dose, it should be dissolved in a few ounces of distilled water with a sufficient quantity of sulphuric acid, and given every hour by the large spoonful. If the stomach cannot bear this, give an injection of a double dose, with the precaution, not to inject more than one ounce of liquid at once, every hour. The action of the medicine will be assisted by friction of quinine ointment along the vertebral column, on the articulations of the wrist, knees, and under the arm-pits, etc.

Some enthusiasts consider sulphate of quinine, as a preventive, and direct it to be taken in a perfect state of health, or administer it in the outset of the fever. I have tried this without having felicitated myself. I will say as much of the association of calomel with sulphate of quinine. This combination should be rejected.

I have nothing to add to what I have already said as to blisters.

Bicarbonate of Soda—Nux Vomica.—When the patient complains of nausea, disposition to vomit, of eructations warm and acid, that he feels in the throat and liver a burning sensation, bicarbonate of soda is the remedy indicated. I give it in doses of one gramme in six ounces of distilled water, taken by the spoonful every hour.

I have stated before, how and under what circumstances nux vomica ought to be given. The effect of these two last remedies is often much more sure, if their action is assisted by cold fomentations upon the abdomen, perhaps with cold water alone, or with camphorated alcohol and belladonna.

Belladonna—Camphor.—Compresses of camphorated alcohol and belladonna, placed upon the epigastric region, diminish the beating of the cœliac trunk, the epigastric pain, and the vomiting. Laid at the bottom of the abdomen, they quiet the colic pains and facilitate the passage of urine. Camphor alternated with belladonna, finds its use internally, in combatting hiccough, and camphor alone is especially useful in the typhoid period of the disease.

Tannin.—Tannin diminishes the excitement of the stomach. I recommend its employment, where nitric acid reveals the commencement of albuminous deposit in the urine. Its use must be suspended, if the albumen persists or increases. Tannin is administered every hour, in grain doses in a spoonful of water. When the twelfth grain has been given, and it works no favorable change, it is replaced by arsenic.

Arsenic.—Towards the end of the second period, when the vomiting cannot be arrested, when the patient has continual nausea, when the vomit contains bile or mucosities filled with blackish or sanguinolent streaks, in a word when the characteristic signs of pronounced yellow fever are developed, there is no better remedy than arsenic. It is given as arsenious acid dissolved in water, and prepared in the following manner: Boil for an hour a grain of arsenious acid in a porcelain cup, containing a half pint of distilled water; then replace the evaporated liquid with an equal volume of boiling water, let it cool, and give this solution by the teaspoonful every half hour, until the nausea and vomiting cease. The administration of this remedy is continued for two days, at longer intervals, that is every hour, then every two hours, finally every four hours. Prescribed under fitting circumstances, arsenic often brings unhoped for amelioration.

There are some medicines, whose action, though certain, is inexplicable. Such is arsenic, the influence of which must be accepted as a fact, without considering theories more or less satisfactory. I should add, that arsenic often determines a deceptive hunger, to which there should be no concession, because at this period of the disease, the lightest broth might cause fatal indigestion.

I have tried every possible remedy for black vomit, and there is not one, which has constantly given the same result. I have had extraordinary success with agents, which at other times produced no effect; and I affirm, that there is no therapeutic agent, which can always be employed with

entire confidence. Black vomit is the symptom of alteration, more or less profound, of the bile and of the blood. If it is alteration of the bile, presenting solely the black color of jet, hope remains; but when the vomited matter is of the color and consistency of coffee grounds, the patient is irretrievably lost. This truth rests on an experience of forty years.

It is not surprising then, that under a great number of circumstances, the most heroic agents are absolutely ineffective.

Iron.—It is not necessary to give iron in too large doses. The two best ferruginous preparations are iron reduced by hydrogen, and the muriatic tincture of perchloride of iron. Small doses, often repeated, are much more easily absorbed than large doses. A quarter of a grain of powder of iron every hour, or a drop of perchloride of iron in three ounces of water, taken by the spoonful, every hour, is all that the stomach can bear; more is rejected by the stool or by vomiting. The reconstructive action of this remedy will be assisted by cold lemonades, and by ice in small quantities. Cold vinegar lotions over the whole body, frictions, enveloping the patient in sheets wet with cool vinegared water, compresses of cold vinegared water on the abdomen, changed as soon as they become warm, are adjuncts, which should not be neglected, and which will always be found good.

Ice.—Ice is one of the agents greatly abused, especially in the first and second period. It is an excellent tonic; but I am not well assured of its employment in the third period.

Drinks.—During the first period, the diet drinks ought to be warm or hot, to facilitate the diaphoresis so necessary at this time. But in the second and third period, there is used only cool water, slightly acidulated, and sweetened *ad gratam saporem*. In the great majority of cases the patient prefers simple water.

Regimen.—Absolute diet is demanded, rigorously, while the fever lasts. But when the febrile symptoms have disappeared entirely, and at the same time local congestions dissipated, a little thin broth may be allowed.

A certain sensation of hunger is felt, especially towards the end of the first period; but the desire of the patient must be resisted, although the pulse maybe less frequent and less full. Often at the first touch, the pulse seems regular, but the attentive physician will find something abnormal, and he will soon be assured, that the improvement is more apparent than real.

Convalescence.—The greatest care should be given to convalescents, because relapse is often fatal. The nourishment ought to be select and the patient should not be exposed to the sun or to the influence of the moon. When the disease does not go beyond the first period, convalescence is much shorter, if there is no leading organ assailed; but if it reaches the second and third period, especially that of hemorrhages and profound alterations of the blood, convalescence is long and painful, and often leaves its traces during the whole life.

When restoration is complete, wine of cinchona, wine of iron, cold baths, and sea baths are prescribed. In cases of swollen parotids, convalescence is prolonged during many months.

Is it possible to prevent a disease, which makes such ravages? Its entire destruction seems to me an unrealizable utopia, because we have seen, that one of the causes of its existence is in the atmosphere: but it is certainly possible to diminish its effects, and to avoid it, when it exists. A well observed hygiene would give the best results, and the government, which would attach its name to this undertaking, would deserve well of mankind.

Besides the causes of the disease, described in the commencement of this memoir, there is one important cause, in the collection of persons living together under the same roof, especially in barracks. These establishments are designed to shelter, day and night, strangers, who expose themselves without precaution to sun and to damp, and who drink, and eat, as if they had nothing to fear. These barracks ought to be built on elevated places, far from the city, and from the seashore, and especially, be well ventilated. The surgeon of the regiment, should have a roll-call three times a day, and, on his responsibility, put in the infirmary, and on diet, any one, who complains of the least headache.

Instead of sending to one hospital only, the sick coming from garrison and from government ships, several military hospitals ought to be established. Every barrack should have its infirmary, with medical service, and the sick should not have to cross the city to obtain treatment. There is no disease which develops contagious miasm more suddenly than yellow fever. A large accumulation of sick in the same place, is a certain source of disease and contagion.

On board of vessels, a good hygienic system would greatly diminish the number of the sick. In ordinary times, government vessels should not have

full crews. The men will then have a sufficient quantity of air to breathe. They should not be drilled in the fierce heat of the sun, or in rain, and they should be required to go to the surgeon, for the least pain in the head.

We have observed, that the incubation of yellow fever, is from ten to twelve days. It will be prudent then, in the heated term, not to remain in port more than six days. English vessels have the habit of not remaining more than three days, but they cruise in the vicinity, or go to Jamaica, so that they are always exposed. The best plan would be to leave the waters of the Gulf, which is the true centre of infection. When, despite every precaution, there are sick men on board, and their condition inspires apprehension, they should be put ashore at once, the vessel whitewashed, and sail set for other latitudes to the north.

Commercial are not under the same conditions as government vessels. On the former all is sacrificed to speculation. The crew is lodged as closely as possible, and ten or twelve men are often seen in a space, where four men can scarcely lie down. The government should require the lodgment of the crew to be on deck, so as to allow free circulation of air. Their food should be wholesome, and well chosen, and instead of giving the crew fresh meat every day, it is better to continue giving them salt meat which is more wholesome, and more nourishing. Let us add, that the meat of the country, bought by merchantmen, is not of first quality.

There are some captains, who, to shun the invasion of the disease, engage blacks to load and unload their vessels, and during this time, leave their crews inactive. This plan is of no avail. The crew is generally composed of strong men, habituated to bodily exercise, which facilitates transpiration. They require, therefore, moderate labor.

The water of the country is bad, and it would be good to add to it a little brandy or rum. This drink is better than wine, or beer, which are adulterated and often give colic.

Prophylactic remedies have been much recommended. In latter times, chlorine on board ships was extolled, and has fallen into disuse, like others I have tried without satisfaction. I have seen a captain, who, convinced of possessing the universal panacea, neglected his sailors, and they became gravely sick. Prophylactics can not have any action on a disease which is in the air. Hygienic precautions and cleanliness, are worth more than these pretended preventives.

Inoculation.—At one time, it was attempted at Havana to prevent yellow fever by inoculation of the poison of a snake, supposed to be the *crotalus horridus*. A German adventurer assumed the respectable name of Humboldt to sustain his theory. This man had observed that the Indian prisoners, lead from Mexico to Vera Cruz, exhibited, when they were bitten by a viper, symptoms analogous to those of yellow fever. He then made some experiments with the inoculation of this substance at Vera Cruz, and at New Orleans. He came afterwards to Havana, and obtained from General Concha, then Governor of the Island, permission to make some trials at the military hospital. He proceeded in this wise: He made, said he (no one saw it), the snake bite the liver of an animal, and kept it to putrefaction. He inoculated with this substance, and gave at the same time, internally, a syrup composed of *mikiana-guaco* and rhubarb, with the addition of iodide of potassium and gutta-gamba.

The symptoms appeared in the following order: at the moment of inoculation, the subject was taken with a transient vertigo, at other times, with a nervous trembling, which lasted a longer time. Seven hours after, the pulse was modified in a permanent manner, more frequent or slower, stronger or weaker. Eleven hours after, he had febrile heat. At the end of fourteen hours, he had headache, inappetence, thirst; sixteen hours after, the face red, the conjunctiva injected. From the outset, the gums were swollen and the patient suffered from colic. Eighteen hours after, the gums were painful, and their borders became red, with pains in the salivary glands, and in the nerves of the face and teeth. Nineteen hours after, pain in the lower jaw, in the direction of the submaxillary nerve. At the twentieth hour, bad taste in the mouth, coryza, and œdema of the face; at the expiration of twenty-two hours, a sensation of contraction of the throat, without visible change. At the twenty-third hour, jaundice; at the twenty-fourth, gingival hemorrhage; at the twenty-eighth, conjunctiva injected, chills; at the twenty-ninth, tonsillary angina; at the thirtieth, pains in the loins; at the thirty-eighth, pain in the joints; at the seventy-second, swelling of the lower lip. During convalescence, prurience of the skin, with cutaneous eruptions. These symptoms are far from being those of yellow fever. They belong in part to those produced by the mixture of guaco and iodide of potassium, and in part to those, which putrifying substances produce, when they are absorbed.

M. Humboldt would not yield to the desire of the Commission, to try solely the inoculation and the syrup of guaco. The conclusions of the report were absolutely unfavorable to the experimentalist. The epidemic of yellow fever continuing its course, the proportion of mortality was the same among the inoculated, as among other subjects, and if the statistical tables, presented by M. Humboldt, disagree with the conclusions of the Commission, it is because he had among the inoculated, not only a great number of acclimated persons, but of individuals who had already had the disease. These ideas of inoculation, inspiring a false security, might produce the saddest results. I think, however, that this interesting question might deserve to be studied anew.

I here terminate my effort. I have proposed to make yellow fever known, as I have observed it at Havana for some twenty years. The reflections, which I have presented on the nature and the causes of this terrible disease, the details into which I have entered, in order that all that relates to its symptoms, its progress, and its treatment, may be clearly estimated, make this work a monograph, the utility and opportuneness of which, will, I hope, be appreciated by those medical men, who are called to practice in tropical regions.

The Elastic Bandage in the Treatment of Aneurisms.—It seems likely that Esmarch's bandage will add very greatly to our means of treating aneurism. Dr. Weir has collected twenty-one cases of iliofemoral, femoral, and popliteal aneurisms, mostly the latter, treated in this way. Twelve of these were successful, while the others failed, owing chiefly to the fact that obstruction to the arterial current was not kept up after the removal of the elastic bandage. Upon this point Dr. Weir lays great stress, and states that in it is the gist of the treatment.

In connection with the study of this matter, the question of how long a limb can be kept desanguinated is of importance. In the lower animals the time is six or eight hours. In man the time is longer than has been heretofore supposed. Ischæmia has been enforced for four, five, and in one case fourteen hours without injury. During the compression it is important to remember that the arterial tension elsewhere is increased.

Autopsies have made it probable that coagulation begins in the tumor and extends up several inches into the artery. The arterial clot then becomes organized into fibrous tissue, and for this organization a healthy state of the wall is necessary. Aneurisms with large mouths are perhaps more easily cured by Esmarch's bandage.

As the result of a study of the cases collected, including his own, Dr. Weir recommends a plan of treatment like the following: the limb should be bandaged up to the tumor and above it, but not over it. The patient should stand erect before the upper bandage is put on. Tubing should be applied in the usual manner. The elastic compression may be kept on for two hours, followed by the application of a tourniquet for two hours. If pulsation is still apparent, the elastic and mechanical compression should be repeated until pulsation has ceased. After consolidation of the tumor is secured it is well to moderate current above the tumor for twelve or twenty-four hours by a bag containing seven or ten pounds of shot.—*Amer. Jour. Med. Sciences*, Jan., 1879.

EDITORIAL.

NORTH CAROLINA MEDICAL JOURNAL.

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M. J. DEROSSET, M. D., New York City, Editors.
THOMAS F. WOOD, M. D., Wilmington, N. C.

 *Original communications are solicited from all parts of the country, and especially from the medical profession of The Carolinas. Articles requiring illustrations can be promptly supplied by previous arrangement with the Editors. Any subscriber can have a specimen number sent free of cost to a friend whose attention he desires to call to our JOURNAL, by sending the address to this office. Prompt remittances from subscribers are absolutely necessary to enable us to maintain our work with vigor and acceptability. All remittances must be made payable to DEROSSET & WOOD, P. O. Box 535, Wilmington, N. C.*

THE APPROACHING MAY MEETINGS.

We ask the attention of the members of the North Carolina Medical Society, at the request of Dr. Charles Duffy, Jr., President, to their obligation as Chairmen and members of Committees and Sections.

The sections as instituted at the last meeting of the Society are as follows:

Surgery and Anatomy.—Dr. Charles J. O'Hagan, Greenville.

Obstetrics Gynæcology.—Dr. H. Otis Hyatt, Kinston.

Practice of Medicine.—Dr. W. A. B. Norcom, Edenton.

Materia Medica and Therapeutics.—Dr. G. G. Smith, Mill Hill.

Microscopy and Pathology.—Dr. G. G. Thomas, Wilmington.

It will be remembered that the resolution creating the Sections offered by Dr. Shaffner, of Salem, designed that all papers coming under the above heads should be presented to the Chairman of that section, and through him papers are to be presented to the Society. It is highly desirable, therefore, that papers intended to be presented should be sent forward to their proper chairman, that they may get early attention. The ultimate design of this method is to promote a systematic and orderly presentation of papers, and to induce every contributor to the literature of the Society to put his paper in a way to be properly shaped before being read to the Society. Sometimes papers are too long to be read during the session, and still are too important to be neglected. These papers should be passed upon by the Section to which they belong, and given to the publication committee.

As desirable as this plan is, it must not be understood to exclude papers prepared too late to be reviewed by the section, for literary laziness and procrastination is the prominent failing of members of our State Medical Society. (We are now speaking as editors).

With the additional work now devolving upon our State Society, every effort will have to be made to economize time. It is desirable that the Board of Medical Examiners meet a day in advance of the Society, in order that candidates passing the Board can at once enter into the duties of full

membership, and enable the members of the Board also, to take active part in the proceedings. This we understand to be the plan agreed upon by the Board, and it may be officially announced in this JOURNAL.

All these matters should be thought over before the meeting is right upon us, if we intend to make the best use of the opportunities presented, and not embarrass the presiding officer by a jumble of ill-digested work, or bring disrepute upon the Society by presenting papers put together without due study.

Another matter of vital importance should be carefully considered by every member of the Society. It is the amendment proposed by Dr. T. D. Haigh, of Fayetteville. He proposes to amend the Constitution (Art. IV, Sec. 2,) so that the officers are elected by ballot. This is not a new feature. It has been tried before in the Society but was found to consume a great deal of time. This is the only objection we have heard against it, and this should not be considered insuperable, if the amendment corrects abuses of which we have heard complaints.

We would like to see the office of President filled for a longer term than one year. A good presiding officer is not so easy to get that we ought to be willing to let him go out of office as soon as he has shown his capacity, and this remark applies with peculiar force to the present incumbent. To affect this change though, there must be a further amendment of the Constitution.

YELLOW FEVER POISON SURVIVES A WINTER.

“The U. S. Steamer Plymouth, Captain Hanning, which left Boston March 15th, for a cruise to the West Indies, returned to Vineyard Sound on account of two cases of yellow fever occurring on board when about 80 miles south-east of Bermuda Islands.

“The ship had been in Boston during the winter, and as she had come from the West Indies last autumn with yellow fever on board she had been frozen out and fumigated. As she had not called into any port since leaving Boston, this development showed that the germs of yellow fever still existed in her, and she was headed north, being deemed, under the circumstances, unfit for cruising in the tropics. On the 31st of March, Peter Eagan, the boatswain’s mate, was buried, having died from yellow fever on the previous day.”—*Wilmington Sun*’s associated press telegram.

The above dispatch has since been verified and the minute details will no doubt be investigated most thoroughly. Notwithstanding this case is not without a parallel, it comes in uncomfortable collision with the theories we cherish of the killing power of low temperature on the yellow fever poison.

In the most dismal times of a ravaging epidemic the heart turned with anxious longings for the arrival of frost! This was the line of demarcation between the pestilence and recovery from it! But in this case we are informed that the Plymouth spent the winter in Boston harbor with open hatches, the cold being intense enough to freeze the water in the boilers. Every means for thorough disinfection had been applied that could suggest itself to the minds of the well educated medical officers in the service of a government lavish in its supplies. With all this, a short cruise develops the fever in a form intense enough to cause the death of one of the two seized with the disease.

We will await the detailed accounts of the investigation which is to follow with peculiar interest. It is a starting point for the National Board of Health, and a difficult one.

We append the following from the Surgeon-General of the Navy, received through the Bulletin of the Public Health, from Surgeon General Hamilton, U. S. M. H. S.:

“The Surgeon-General of the U. S. Navy has furnished the following facts in regard to the recent outbreak of yellow fever on the U. S. Steamer ‘Plymouth:’ On November 7th last, four cases of yellow fever occurred on

board the vessel while lying in the harbor of Santa Cruz; these were removed to hospital on shore and the ship sailed for Norfolk. Three mild cases occurred during the voyage and the 'Plymouth' was ordered to Portsmouth, N. H., thence to Boston. At the latter port everything was removed from the ship and all parts of the interior freely exposed to a temperature which frequently fell below zero, the exposure continuing for more than a month. During this time the water in the tanks, bilges, and in vessels placed in the store rooms was frozen, 100 pounds of sulphur was burned below decks, this fumigation continuing for two days, and the berth-decks, holds and store rooms were thoroughly whitewashed. On March 15th, the ship sailed from Boston southward; on the 19th, during a severe gale, the hatches had to be battened down, and the berth deck became very close and damp. On the 23d two men showed decided symptoms of yellow fever, and on the recommendation of the Surgeon, the vessel headed northward. The sick men were isolated, and measures adopted for improving the hygienic condition of the vessel and crew. The surgeon reported that he believed the infection to be confined to the hull of the ship, especially to the unsound wood about the berth deck, all the cases but one having occurred within a limited area, and that while the 'Plymouth' is in good sanitary condition for service in temperate climates, should she be sent to a tropical station, probably no precautionary measures whatever, would avail to prevent an outbreak of yellow fever."

Charcoal for Burns.—A retired foundryman claims that powdered pine charcoal thickly dusted over a burn is a never-failing and speedy remedy.

THE NATIONAL BOARD OF HEALTH.

This body as now composed includes fairly representative men. As far as we can learn it is as follows:

Dr. James L. Cabell, University of Virginia, President.

Dr. John S. Billings, U. S. A., Washington, Vice-President.

Dr. Henry J. Bowditch, Boston, Mass.

Dr. Henry A. Johnson, Chicago, Illinois.

Solicitor-General, Samuel Phillips, North Carolina.

Dr. S. M. Bemiss, New Orleans.

Dr. Th. Turner, Surgeon U. S. N., Washington, Secretary.

Dr. P. H. Bailhache, U. S. Marine Hospital.

Dr. Robert W. Mitchell, Memphis, Tenn.

A committee of experts has been sent to Havana to study the disease where it is endemic, and where it can be seen for many months in the year.

“The system [adopted by the new National Health bill] contemplates a national sanitary supervision of all vessels engaged in the transportation of goods or persons from any foreign port where any contagious or infectious disease exists, to any port of the United States. All such vessels shall be required to obtain from the consul, vice-consul, or other consular officer of the United States at the port of departure, a certificate in duplicate, setting forth that said vessel has complied with all the necessary regulations and possesses a clean bill of health. This provision applies with particular and special force to vessels from Havana, a clause in the bill defining in detail the duties of the medical officer in charge of the port. The said inspector must issue a certificate setting forth ‘that he has personally inspected said vessel, her cargo, crew, and passengers; that the rules and regulations prescribed by the National Board of Health in respect thereto have been fully complied with, and that in his opinion the said vessel may be allowed to enter any port of the United States and land its cargo and passengers

without danger to the health thereof on account of any contagious or infectious disease.’ Any vessels from such port entering any port of the United States without such certificate shall in each instance forfeit the sum of five hundred dollars. The execution of these provisions is entrusted to the National Board of Health. The latter is also charged with the duty of obtaining information of the sanitary condition of foreign ports and places from which contagious diseases are or may be imported into the United States, and also similar information from home ports. It is also provided that the National Board of Health ‘shall correspond with similar local officers, boards and authorities acting under laws of the States in sanitary matters, to prevent the introduction and spread of contagious and infectious diseases from foreign countries into the United States and from one State into any other State by means of commercial intercourse, or upon and along the lines of inter-State trade and travel.’ To such an end it shall be lawful in times of emergency for said board of health to confer upon any such local officer or board within or near the locality where his provisions of this act, and any rules or regulations made in pursuance thereof.”—*Medical Record*.

SYMES ON THYMOL AND THYMOL-CAMPHOR.

Dr. Symes, in the *Pharmaceutical Journal* of January 10, publishes the results of his researches on the combination of thymol, chloral-hydrate, and camphor, acting as an antiseptic. The two former drugs are rubbed together in a mortar, and an equal quantity of camphor added, which liquefies the whole, and produces a powerful antiseptic. Its virtues were immediately tested on some urine containing pus, and which was already beginning to decompose. Two drops of the compound being added to it, the putrefaction was arrested. If thymol and camphor alone are rubbed together, they also become liquid, and this a convenient form from which to prepare the ointment. Thymol-camphor can be mixed in almost any proportion with vaseline, *ung. petrolei*, or ozokerine, and the thymol will not separate, as in crystals, when thymol alone is used. A solution of thymol in water (1 in 1000) is sufficiently strong for the spray in surgical operations. If used for the throat, milk and glacial acetic acid will be found to be good solvents for it.—*London Medical Record*.

REVIEWS AND BOOK NOTICES.

MODERN SURGICAL THERAPEUTICS: A Compendium of Current Formulæ, Approved Dressings, and Specific Methods for the Treatment of Surgical Diseases and Injuries. By George H. Napheys, A. M., M. D., etc. Sixth Edition. Revised to the most recent date. Philadelphia: D. G. Brinton, 115 South Seventh Street. 1879. Pp. 605. Price \$4.00, in cloth.

This is a companion volume to Napheys' Medical Therapeutics which we noticed in our January issue.

The design of this work is to give a careful digest of surgical therapeutics up to the latest date, and the author has succeeded in carrying it out. As a work of ready reference it may be compared favorably with any of a similar character. Discrimination in selections, however, does not seem to be the aim of the author, but rather to bring all matters under their heads, leaving the reader to select those best suited to his needs.

In divesting surgery of its operative procedures, it leaves a comparatively indifferent number of resources, but the therapeutical branch is by no means at a stand still.

We are pleased to see that under the head of anæsthetics, chloroform has been allowed its proper place at the head of the list.

Chloroform "is the most potent of all anæsthetics," he says, "and its use is still advocated by many eminent surgeons. Only the alleged dangers attending it, prevent its exclusive employment. Many of these arise from its ignorant or heedless administration." The directions for its use are given, as also the means of combatting dangers arising from it. Dr. Napheys might have added with a great deal of truth, that chloroform should not be administered by any surgeon who is not habitually on his guard as to the dangers of the anæsthetic state.

The dressing of wounds after the new processes of antiseptic practice receives a great deal of attention. To one familiar with the dressings during our civil war, on examination of the present multitudinous plans to exclude “germs” would bring back the days of our grand-fathers in surgery with their balms and balsams and salves; and some of the dressing is not more rational. According to Esmarch (p. 151 and 152) the dressing of gun-shot wounds should be purely antiseptic. “Do not examine the wound at all, rather than examine it with unclean fingers—and everything is unclean, in the strict sense that is not antiseptic.

“* * * * To avoid pernicious putrefactive influences the wounds must not be touched by the hands, but closed rapidly by antiseptic plugs, in order to preserve them from the contact of putrefactive agents until they can undergo the Lister treatment in the hospitals if necessary. He proposes that every soldier should carry in the lining of his uniform two balls of *salicylated jute* wrapped in gauze.”

We make this particular quotation to show to what old-maidish precision the antiseptic idea is leading good surgeons. This ever-present inextinguishable “germ” is the evil spirit hovering over every wound. Nets of gauze are set to protect it; strong odors from the witches cauldron are summoned to stifle and destroy the malicious fiend.

We are thankful though that the civil surgeon still sees “union by adhesion,” and “first intention,” and “granulation,” in regions so far remote from Listerism that there is little hope it will enter there, and if it does it will hardly captivate the even-minded country surgeon. When the days of probationary Listerism have ended, we will not be surprised if the verdict is against it.

But we have digressed from our book. It is the XVIth chapter on “Diseases of the Skin” that will be often consulted by the busy doctor. Having made his diagnosis, here is a goodly array of remedial agents, from the most eminent teachers to help him out of difficulties. We miss chrysophanic acid in the composition of his formulæ for the treatment of psoriasis. It certainly has made as much headway in the favor of the general practitioner as any of the more recent agents.

But why say anything about a book which has made its way through the world, and has now come to its sixth edition? The hundreds of medical men who will read it, will traverse a field of surgical treatment far beyond the

facilities of those possessed of the best private libraries. As long as the author keeps up with the current of surgical treatment, his book will be sought after. We congratulate the author, and Dr. Brinton, on the success of this book, and advise our friends to buy it.

A CLINICAL TREATISE ON DISEASES OF THE LIVER. By Fried-Theod. Frerichs. Prof. of Clin. Med. Uni. of Berlin, &c., &c. In three volumes. Translated by Charles Murchison, M. D., F. R. C. P. Physician to the London Fever Hospital. New York: Wm. Wood & Co. 27 Great Jones Street. 1879. 8vo. Pp. 224.

This is the third volume of Wood's Library of Standard Medical Authors.

For many years this work of Frerich's has been a classic, although only known popularly to the American profession by the large number of quotations made from it by writers on diseases of the liver. Although the word "Clinical" appears on the title page, it is nevertheless a systematic treatise which traverses the entire field of clinical pathology, and embraces also lucid historical accounts of the phases of change which medical men have passed through on their way to the knowledge of the present day.

This book, more than any we have been called upon to review, shows how much German authors rely upon the authority of their own people. References everywhere abound, but for the most part to German works. We do not mention this as a fault, but to make the contrast with American authors who seem to glory in going far away from home for authority among the unspeakable names of the Russian and German gentry.

Prof. Frerich's work for this reason will be more valuable to American students who wish to know the state of pathology in Germany in regard to "the great gland."

As, of course, no American physician can now forego the pleasure and duty of making Frerich's on the Liver one of his working tools, we leave them to judge if we are mistaken in saying that it is a master-piece.

Dr. Charles Murchison is the translator, and his preface serves to elucidate many points, and to bring the work up to the present advanced state of pathological and physiological knowledge. It is not necessary to say anything commendatory of the author of Functional Diseases of the Liver,

as every Southern physician will have found a good friend and counsellor in this volume already.

To subscribers this work is sold at \$1.00, a very low price!

THE DISEASES OF LIVE STOCK and their most efficient remedies: Including Horses, Cattle, Sheep and Swine. By Lloyd V. Teller, M. D. Philadelphia: D. G. Brinton, 115 South Seventh St. 1879. Pp. 469. Price \$2.50.

Diseases of the domestic animals deserve more study from the medical profession than they receive. Medical men even now submit their horses to the treatment of the neighborhood blacksmith and farrier, whose ignorance and brutality is all but universal, rather than inform themselves of the phenomena of brute diseases; in fact, some medical men hold it as beneath their dignified calling to give their attention to such affairs. Fortunately now a better day is dawning, and books like this will do a great deal towards enticing physicians into this neglected field. There is no practice that promises such profitable returns as the educated and skillful management of diseases of domestic animals.

We advise our friends in the country to put this volume side by side on their book-shelves with Youatt, and soon the latter would be but a shelf-keeper alongside their new acquaintance.

The point of view from which the study of the diseases of domestic animals is growing in importance, is the relation of their diseases to ours.

To be able to detect measly beef and mutton is an accomplishment that every physician should acquire, now that we know that tape-worm has its origin there. And we should also be stimulated to earnest enquiry when we remember what great results JENNER brought out of the study of cow-pox.

BIENNIAL REPORT OF THE NORTH CAROLINA INSTITUTION FOR THE DEAF AND DUMB AND THE BLIND. From January 1st, 1877 to January 1879. 32d and 33d Sessions. Raleigh: Published by order of the Board of Trustees.

The last Legislature was famous for fault-finding, but had nothing but praise for the Institution presided over by Mr. Gudger. His report shows

good work done, and common sense ideas of the theories of the methods of teaching of those unfortunates under his care.

Mr. Gudger reviews the arguments of the advocates of the Manual method, and the Articulation method, of instructing deaf mutes as follows:

“There is a ground, however, upon which the advocates of each system can meet and agree. In most of the larger institutions articulation has been introduced and is a success, when the class to be instructed consists of those who, having heard in childhood and learnt to speak, have become deaf (and so are in danger of losing what speech they have) or of those who are partly deaf and consequently not able to catch the delicate shades of sound in different words similar to each other. As these persons have some language to build upon, and an idea of sound, it is comparatively easy, by means of the skillful methods in use, to improve and advance their knowledge in this particular, especially as the teacher may use the *known* in getting at the *unknown*; but to attempt to teach articulation to an ordinary congenital deaf-mute, is to spend valuable time in that which gives promise of little fruit. In other words, as our matter-of-fact American people would express it, ‘*It does not pay.*’”

The entire report shows that the management is in the hands of a courageous and enthusiastic worker,—one not too much engrossed with the beautiful theories of his profession—who shows practical results instead of learned dissertations.

LECTURES ON PRACTICAL SURGERY. By H. H. Toland, M. D. Professor of the Principles and Practice of Surgery, &c., &c., in the University of California. Second Edition. Illustrated. Philadelphia: Lindsay & Blakiston. 1879.

This is a handsome volume of 520 pages, written by a teacher of surgery of great celebrity on the Pacific slope. It consists of lectures as delivered in the Medical College of the University of California, reported by a stenographer. The first edition of this book, although it was treated rather severely by the critics has found ready sale, the present being the second edition.

It is not difficult to see that Dr. Toland is an original teacher of merit, bound down by no school, nor easily captivated by innovations. He is

confident of his powers and does not speak with uncertain meaning.

Under the head of fractures of the thigh, the apparatus in favor with the author is the double inclined plane with some modifications, and with which he has had admirable results.

“When you engage in practice,” he says p. 284, “you will soon be convinced that the double inclined plane and short splints are generally better than a more complicated apparatus.”

Again—“If physicians relied more on their common sense than on the rules of authorities in the treatment of fractures, there would not be so many cases of deformity resulting from such injuries as are daily presented. I would as soon think of committing suicide as of placing an oblique fracture of the tibia in an ordinary fracture-box, filled with either sand, sawdust, or any of the other substances used for that purpose,” p. 279.

A case of aneurism of the left iliac artery is given and illustrated, (p. 515). “In aneurism of the external iliac artery” the author says “I never open the sheath, and consequently apply a single ligature; the sheath of the vessel not being disturbed, there is scarcely a possibility of the occurrence of secondary hemorrhages. I have ligated the external iliac nine times, and my success is the best evidence of the correctness of the theory upon which it is based. One patient died from gangrene of the extremity, and the other from internal hemorrhage which proceeded from the small vessels that were lacerated when the peritoneum was detached from the iliac fossa,” p. 516.

The volume is well illustrated by fresh designs,—all of them original—a matter of sincere congratulation to the author and publishers.

A book possessed of so much originality and individuality as this, will be sure to find a large number of readers among the former students of the author, and will also make its way into favor with the student of American surgery.

Syphilis by Vaccination with Human Virus.—The virus was taken from the arm of a child aged seven months, apparently in perfect health. Twenty-five girls were vaccinated from this infant. At the end of six weeks, twelve of the girls were taken with symptoms of syphilis, ulcerations at point of inoculation followed by exanthema, ulcerations in mouth and pharynx, condylomata of anus, syphilitic ozæna, etc.; three others of the group suffered from suspicious ulceration near the vaccine sore, which failed to be followed by constitutional symptoms. Later it was discovered that the mother of the child was suffering from syphilis.—*Hosp. Med. Gazette.*—*Louisville Med. News.*

NEW JOURNALS.

INDEX MEDICUS.—We hardly know which to admire most in this new journal—its typographical excellence, or its editorial management. It is no surprise to the medical public that it begins its existence as an accomplished success, as Dr. Billings had long ago shown his capacity in his official position as librarian, and has shown his taste also in the typographical selections in the specimen fasciculus of the catalogue of the National Medical Library.

The two numbers now before us demonstrate as it could be done in no other way, the necessity for some guide to the medical literature of the world. The student will be helped in pursuing any special research, and the general reader will be able to know what is going on in the medical world, and be saved very much irksome reading by following his inclinations. The careless and uninformed “discoverer” of new things, may be saved the trouble of re-discovering, by looking out into the field upon what others are doing.

Twice we have written notices of this periodical, and each time it was overlooked. Our notice though tardy is none the less earnest.

The subscription price is \$3.00 a year, and intending subscribers should commence at once with the first number. Address, F. Leypoldt, 37 Park Row, N. Y.

THE COURIER OF MEDICINE AND COLLATERAL SCIENCES is the title of a new monthly journal of exceptional excellence, commencing its career with the January number, in St. Louis. This city had already in the field a brilliant array of medical journals, and while we cannot think any less of our old friends we welcome the new one.

The printer's art has cleverly served up the literary matter in a style which will commend itself to all readers, and especially those who are beginning to hold their books at arm's length. The effigy of John Hunter on the outside cover is a faithful copy of the celebrated Sharpe engraving, and is in good taste.

What a clever faculty there must be in St. Louis to sustain so many good journals by purse and pen! We wish for the journal a most hearty

appreciation.

NORTH CAROLINA FARMER.—An examination of the April number of this periodical was a pleasant surprise. It abounds in practical matters suited to the necessity of our farmers, and should be read and supported by the pen and pockets of the entire agricultural community.

We make one suggestion to the editors, and that is, that if they are to have a column for diseases and remedies (a questionable matter for all non-medical publications), that it should be in the hands of a competent medical man. We congratulate the publishers that they have no nostrum advertisements. Large quarto of 18 pp. at \$1.00 a year. Jas. H. Enniss, Editor and Publisher, Raleigh, N. C.

GROSS ON THE TREATMENT OF CYSTIC GOITRE.

In a clinical lecture delivered by M. Gross, of Nancy, reported in the *Revue Médicale de l'Est*, of November 15, he describes the treatment of cystic goitre, known as Michel's "mixed method," as extremely useful, and furnishes a case illustrating its advantages. Giving a rather extended review of the various modes hitherto proposed for removal of these growths, he points out their drawbacks, and the superiority of Michel's method over them. Briefly the latter consists in making a vertical incision in the skin over the most prominent cyst, and then dissecting carefully down through the various structures, until the wall of the cavity is reached. A very fine trocar is then pushed into the cavity with a canula, and through the latter the fluid is withdrawn. After this a plaque of *pâte de Canquoin*, about three centimètres broad, is applied to the cyst, the sides of the wound being protected by a circular piece of diachylon. This is left on a day or two until an eschar is formed, which soon after comes away, leaving a free opening through, which the cyst can discharge, until it shrinks up, after suppurating for a time.

It is claimed for this method that it is less likely to give rise to dangerous hemorrhage than several others, while, the caustic only being applied to the surface of the cyst, severe inflammation of the tissues around is avoided. Other cysts, if present, are similarly treated through the aperture of the first.
—ARTHUR E. BARKER, in *London Medical Record*.

CURRENT LITERATURE.

REMEDIAL AND FATAL EFFECTS OF CHLORATE OF POTASSA.

In a paper read before the Medical Society of the State of New York, (*Medical Record*, March 5th), Dr. Jacobi reviews in a very careful manner the remedial effects of chlorate of potash, and calls attention also to what he considers the dangers of large dosages so commonly employed by physicians and patients.

Sir James Y. Simpson, introduced chlorate of potash on the theoretical ground of its employment in chemistry to develop oxygen, to supply oxygen to the blood on the part of the fœtus in cases of placentitis.

Many years ago, Isambert and Honie, found chlorate of potash eliminated without any change, and in large quantities, even as much as 95 or 99 per cent. of the amount administered, in the various secretions of the body; that is in the urine, the saliva, the tears, the perspiration, the bile, and now and then even in the milk; no oxygen was developed at all. The theory of Simpson was long ago given up, because it was found out that the same redness was produced in the blood by other alkalies.

Its principle value consists in its effect upon catarrhal and follicular stomatitis; further, in mercurial stomatitis, the former being a frequent and the latter a rare disease in infancy and childhood.

“In regard to [the employment of chlorate of potash] diphtheria, I can give [my position] in a few words. It is this: that chlorate of potassa is a valuable remedy in diphtheria, but that it is not *the* remedy for diphtheria. There are very few cases of diphtheria which do not exhibit larger surfaces of either pharyngitis or stomatitis than of diphtheritic exudation.”

There are also a number of cases of stomatitis and pharyngitis, during every epidemic of diphtheria, which must be referred to the epidemic, perhaps as introductory stages, but which still do not show the characteristic symptoms of the disease. * * * *

The dose of chlorate of potassa for a child two or three years old should not be larger than half a drachm in twenty-four hours. A baby of one year or less should not take more than one scruple a day. The dose for an adult

should not be more than a drachm and one-half, or at most two drachms, in the course of twenty-four hours.

The general effect might be obtained by the use of occasional larger doses; but it is best not to strain the eliminating powers of the system. The local effect cannot be obtained with occasional doses, but only by doses so frequently repeated that the remedy is in almost constant contact with the diseased surface. Thus the dose, to produce the local effect should be very small and frequently administered. It is better that the daily quantity of twenty grains should be given in fifty or sixty doses than in eight or ten: that is, the solution should be weak, and a drachm or a half-drachm of such solution can be given every hour or every half hour, or every fifteen or twenty minutes, care being taken that no water is given soon after the remedy has been administered for obvious reasons. He referred to these facts with so much emphasis because of late an attempt has been made to introduce chlorate of potassa as the main remedy in bad cases of diphtheria—and, what is worst, in large doses.

As early as 1860, Dr. Jacobi advised strongly against the use of large doses of chlorate of potassa. * * * * The treatment is dangerous and because of the largeness of the dose of the chlorate given.

After reviewing the opinions of several writers who have extolled chlorate of potassa in large doses, and having pointed out the real solution of so many having succumbed to nephritis or similar symptoms, he concludes:

“The practical point I wish to make is this, that chlorate of potassa is by no means an indifferent remedy; that it can prove and has proved dangerous and fatal in a number of instances, producing one of the most dangerous diseases—acute nephritis. We are not very careful in regard to doses of alkalies in general, but in regard to the chlorate we ought to be very particular. The more so as the drug, from its well-known either authentic or alleged effects, has arisen or descended into the ranks of popular medicines. Chlorate of potassa or soda is used perhaps more than any other drug I am aware of. Its doses in domestic administration are not weighed but estimated; it is not bought by the drachm or ounce; but the ten or twenty cents worth. It is given indiscriminately to young and old, for days or even weeks, for the public are more given to *taking hold* of a remedy than to *heed warnings*, and the profession are no better in many respects. Besides,

it has appeared to me, acute nephritis is a much more frequent occurrence now than it was twenty years ago. Chronic nephritis is certainly met with much oftener than formerly, and I know that many a death certificate ought to bear the inscription of nephritis instead of meningitis, convulsions or acute pulmonary œdema. Why is that? Partly, assuredly, because for twenty years past diphtheria has given rise to numerous cases of nephritis; partly however, I am afraid, because of the recklessness with which chlorate of potassa has become a popular remedy. Having often met medical men unaware of the possible dangers connected with the indiscriminate use of chlorate of potassa or soda, I thought this Society would excuse my bringing up this subject. It may appear trifling, but you who deal with individual lives, which often are lost or recovered by trifles, will understand that I was anxious to impress the dangers of an important and popular drug on my colleagues, and through them on the public.”

DEMANGE ON AZOTURIA.

The importance of the study of urology has of late been more fully realized by Medical practitioners, and M. Demange in his thesis (*Thèse de Paris*, 1878) has undertaken to give a full account of the progress of this branch of medical science, being also fortunate enough to be able to enrich it by several new or very little known observations on azoturia. The latter seem the most interesting part of his thesis; we give them here briefly. The normal quantity of urea which must be contained in the urine in the space of twenty-four hours is from nineteen to fourteen grammes. If more or less is excreted, this is caused either by some local or general affection. Some years ago, Bouchard, in studying the causes of loss of flesh in patients suffering from diabetes insipidus, discovered that a large number among them lost an enormous quantity of urea. Having thoroughly examined their symptoms he thought himself fully justified in describing azoturia as a special disease, having peculiar clinical symptoms. The affection begins with a sensation of ravenous hunger, polydipsia or profuse sweating. The thirst is excessive, and the urine passed is generally in proportion with the quantity of drink swallowed by the patient. Its density is from 1000 to 1002. In order to be able to calculate justly the amount of urea lost in twenty-four hours, all the urine passed in twenty-four hours, all the urine passed in this time must be kept and mixed. In some cases it has reached the amount of eighty-seven grammes, a most enormous quantity, which proves that nutrition is very much impaired. Senator Kien and M. Bouchard have shown that what is called extractive matter is eliminated, corresponding to urea in such cases, and that chlorates and phosphates are ejected in a similar proportion. We must, therefore, not be astonished if the patients present general symptoms which are analogous to those of diabetic patients, with the exception of the visual troubles of the latter. Both the crystalline lens and the retina remain intact, and the sight is only influenced by the anæmic state of the brain, which is caused by the dyscrasia, and which in certain cases produces a torpor of the intellect verging on imbecility. As in cases of diabetes mellitus and albuminuria, sometimes the quantity of urea decreases, and even falls below the normal amount.

In order to be able to make an exact diagnosis, it is necessary to examine carefully, both the urea and the other excreta, for several days consecutively. As a rule, persons attacked by simple polyuria, or who are suffering from interstitial nephritis, beginning with polyuria, do not present the symptoms which we have enumerated.

Disturbances of the nervous symptom and alcoholism claim the first place in the etiology of this affection, and indicate the treatment which has to be adopted. It consists in administering drugs to calm the nervous erethism (opium and valerian), and to put a stop to the excessive and progressive impoverishment of the tissues (arsenic, a suitable diet, etc.) Valerian has proved specially successful in different cases, even effecting a complete cure. Besides these cases of azoturia, combined with polydipsia, Bouchard thought that there was another form of the same disease, in which there was no abnormal excretion of urine, although the latter contained an excessive amount of urea. However, as his observations in that respect are far from being satisfactory, and as these are evidently cases of cachexia, the etiology of which is very obscure, it will be wiser to leave them alone for the present. The author then goes on to consider the much-debated question on the varying amount of urea in glycosuria. In some patients suffering from the latter affection, as much urea is eliminated as the general amount in azoturic patients. It is true, however, that there may be something more than a simple coincidence between these two affections, and several authors have tried to link them together. Lécorché, who admits the hepatic theory of the formation of urea, thinks that this is only the double result of hyperactivity of the functions of the liver. Bouchard, on the contrary, considers it as a true complication of the existing affection, where troubles of nutrition are added to those resulting from insufficient respiratory combustion. According to him, the difference between melitæmia and azotæmia consists in the first resulting from the accumulation of a product of secretion, while the second results from the accumulation of a product of secretion. Azoturia is, therefore, as we said, only a complication, an accessory element which must, however, be considered as being a most important prognostic symptom. According to the same author the abundance of sugar in diabetes is owing more to a want of combustion than an exaggerated production of this substance in the organisms. If this be the case, how can we explain the coincidence of an abnormally low temperature with the production of an exaggerated quantity of urea, such as has been observed in every case

without exception? This is the weak point of M. Bouchard's theory, and it would perhaps be better to refrain from giving a decided opinion on the subject until it has been more thoroughly studied. In short, whenever there is an excessive excretion of urea we may consider it as a symptom of incipient cachexia, followed by loss of flesh. The most important question, however, for the medical practitioner is the following: are these two affections to be considered as belonging to two different groups, but having been developed incidentally at the same time in the same patient; or are they connected through a link which is still unknown to us, thereby forming one affection or disease? If these questions could be solved, there might be some hope of discovering some rational mode of treatment, so as to prevent albuminuria from setting in, in which case all is lost. In another chapter we find the calculation of the amount of urea excreted in several chronic diseases, such as obesity, syphilis and athrepsy. Here it is easy to make a mistake, and still more so to err in trying to interpret the results obtained, because here the nourishment taken by the patient plays an important part, which is easily overlooked, e. g., in fleshy persons. Azoturia may be produced either by excess of food, or by abstaining from farinaceous food. The only way of ascertaining if the combusive functions are really exaggerated in a patient would be to compare the amount of chlorates contained in the secretions with the weight of the patient. Since Brouardel published his paper, on what he termed the uropoietic functions of the liver, several experiments have been made to ascertain the amount of urea excreted in diseases of this organ. The results have been very contradictory, but it is certain that large quantities of urea have been found in the urine of patients whose liver was completely degenerated.—*London Medical Record*.

EXPERIMENTS CONCERNING ERYSIPELAS.

Observations “were undertaken” by H. Fillman, of Leipzig, “in order, if possible to obtain further and more accurate information upon some contested points regarding erysipelas.” The experimenter has addressed himself here, especially to the answering of the four following questions:

I. Is it possible to convey erysipelas by inoculation from a diseased to a healthy individual? In other words, do those fluids obtained from the tissues of an erysipelatous part and employed for inoculation (*e. g.*, lymph, blood, the contents of bullæ, pus, etc.,) exercise a specific, *i. e.*, contagious action on healthy individuals when inoculated, or do they not?

II. What is the action of carbolic acid upon those erysipelatous animal fluids which produced the same disease on being inoculated into healthy individuals, and therefore in all probability contained the erysipelas poison?

III. In what way is it possible (apart from direct conveyance) to produce erysipelas in healthy animals by the application of different morbid matters?

IV. What do we learn from the results of anatomical and experimental investigation regarding the presence and significance of bacteria? What relation have they to erysipelas?

It would be difficult for four more interesting or important questions for the practical and scientific surgeon than these. But in proportion to their importance are the difficulties which surround them. These, however, are grappled with by the author courageously, patiently, and honestly, and the result is at all events a series of experiments of extreme interest, whatever be the conclusion, we may feel disposed to draw from them. Indeed, the writer himself seems to have set out upon his investigation purely with the desire to learn whatever is to be learned on the points stated, by careful and patient anatomical research, and without being wedded to any particular theory in regard to them, or desiring to force any conclusion.

Recognizing the great importance of the subject, and the efforts that have been made by others in the same direction to throw light upon it, notably by William, Ponfick, Orth, Bellien, Zuelzer, and Lukomsky, he has recognized many points in which these observers have failed, and has endeavored, in

following out somewhat similar lines of research, to avoid their, and other, errors.

To the danger of one great source of possible error the author seems specially alive, the introduction of other matters into the system of the animal operated on than the mere morbid fluid inoculated, and this he has endeavored to guard against by the most scrupulous cleanliness in obtaining, preparing and introducing such fluids into the bodies of animals.

In touching upon the first question his first case is to define as clearly as possible what are the clinical features which characterize erysipelas in the human subject. He then details in all his experiments, and, from the kind of success of five out of twenty-five inoculations he believes there can be no doubt “that erysipelas is inoculable in rare cases; that fluids from an erysipelatous part, display a specific contagious action.” In three cases he inoculated animals from the human subject successfully with erysipelas, and in two cases animals were infected from other animals. He believes, too, that one human subject might be inoculated from another.

In regard to the second question propounded, four experiments were made with erysipelatous inoculation material, which had been potent in other cases, but here a portion of 2–4 per cent. solution of carbolic acid was added. In none of these cases was there any appearance of either local or general symptoms of any disease.

In answer to question III, all the author’s results were negative. In no case was erysipelas produced by even the most putrid inoculations, when they were not taken from an erysipelatous part. In several cases, however, the animals died of distinct septicæmia.

The observations on the last point which are related in detail, point to the conclusion that bacteria are present in some cases of erysipelas and absent in others, so that we may infer that the advance of the disease does not depend upon their presence.—*London Medical Record*.

CHOLERA INFANTUM.

A contribution to the etiology, pathology, and therapeutics of cholera infantum,^[4] by Dr. T. Clarke, Miller, opens fairly and clearly a theme which will be uppermost in the thoughts of physicians in the approaching hot weather.

The writer begins by pointing out how differently the name *Cholera Infantum* has been applied, including every phase of choleraic diseases. But, “Classification of these diseases to be practical, must of necessity be rather coarse in order to adopt itself to the grain of the great mass—the rank and file—who in the main observe well, though not so systematically as we could wish. The great office-workers do not contribute largely to our mortality statistics, but we will derive great comfort as we proceed, in finding that the figures of these common men are stupendously significant—that the bullet and bayonet are in the aggregate little less important than the epaulette and the gold lace.” He supposed that in the large majority of cases reported as cholera infantum that the choleraic feature was present at some time during the illness, though very likely not at or very near the time of death. For these reasons he considers that the statistics presented are not materially impaired.

“The onset of cholera infantum is characterized by copious watery evacuations from the bowels, often attended by nausea and free vomiting. Attending upon this or even sometimes preceding it, or rapidly succeeding upon it, is the extreme muscular prostration and great depression of the respiratory functions; there is generally more or less griping pain and restlessness, and a rapid appearance of all the symptoms of collapse, coldness of the surface and tongue, feeble rapid pulse, and partial or total loss of voice. Cholera infantum proper, lasts but a few hours—hardly a few days—when it ends in recovery, death, or inflammatory disease of some portion of the intestinal tract; in the latter case the choleraic disease is rapidly rekindled by conditions similar to those which brought about the first attack.

“The condition under which cholera infantum appears, and the *only* condition essential to its development, is continued high temperature day

and night—a mean thermometer above 75°, with small daily range. This high and slightly varying temperature continued from six to ten days, will invariably, in our climate, bring cholera infantum (together with the bowel trouble symptomatically more or less distinct, but pathologically akin to cholera infantum), and the longer this condition of things continues the more numerous and the more intractable the cases become. This is as true in the country as in the city, though we are led to think, as we read the books, that this is a disease of the city especially. Deaths are registered, to be sure, and the books are made in the cities, but if the conditions above mentioned exist in the country, the disease appears there—of course, not a great many cases, for the susceptible bodies furnished by a single block in the city would outnumber those of two or three square miles in the country—yet I am glad to admit that the *conditions* for obvious reasons are not so likely to be present in the country; the contrast, in point of green grass, shade, cool water, and moving air, is no less marked than is the percentage of mortality, and it is *no more* marked.

“Few, if any, recoveries take place until the temperature falls; this fall is usually attended by rain; but this does not seem to be essential, the fall of temperature alone being sufficient to bring about a better state of things. When the temperature falls, cases improve and new cases cease to appear. Sewer emanations do not seem to have anything directly to do with the production of disease, except so far as they tend to impair the general health, and thus diminish the power of the system to resist any disease producing influence, and sewer poisons are no less abundant and deadly at other times than they are when cholera infantum cases are most numerous, and this is the time of year above others when the sewers are abundantly ‘flushed.’”

Filth he does not consider is any more abundant in the summer, and the disease declines with the increase of the very rains that favor increased decomposition. He does not attach much importance to unripe fruits as a causative element. “For whoever saw a youngster who would not exchange all his earthly possessions for a green apple, and whoever saw a child in good health injured by an unripe apple or by any quantity not altogether unreasonable?” And then the sufferers from cholera infantum are all under two years, and hence have not arrived at the green-fruit eating stage of their existence.

Nine tables are constructed, showing the weekly mortality from cholera infantum in Philadelphia, New York, Boston, Baltimore, Cincinnati and Chicago, and also as far as possible the record of thermometric range. An analysis of the table bears out the author's views as regards the causative influence of continued high temperature.

He thinks there are some points of striking resemblance between this disease and sunstroke, so much as to suggest a pathological relationship. 1st. The same conditions seem to be sufficient for and essential to the development of each. 2. They come and go together. 3. The development is gradual and the recovery is slow in each, showing a profound impression made on the living-power of the patient. 4. The explosive character of the attack under the cumulative effects of continued high temperature with the sudden severe or fatal prostration consequent.

"Wash your children well with *cold water* twice a day, and *oftener* in the hot season," is the direction of the New York Board of Health, and Dr. Miller thinks if this one prescription were carried out, cholera infantum cases could be well nigh eliminated from the mortality reports. He thinks it worth while to inculcate among our patrons that however important it may be to take special care in feeding, this will not be sufficient alone, to carry the infant safely through the high temperature of July and August; and we would try to have the people study to keep the little ones *cool*, and the means recommended is cool-bathing or cool-sponging. Medicines are not of much use if the surroundings are cool.

SOCIETY MATTERS.

The subject selected by the Medical Society at the last meeting in Goldsborough, was SPONDYLITIS. They were fortunate in selecting Dr. M. Whitehead as the essayist. It seems to us it would be more agreeable to the essayist, to allow him to select his own theme, and provided he announced it in advance of the meeting, it would answer the same purpose.

The annual oration will be delivered by Dr. W. W. Lane, of Wilmington, upon a subject not announced.

The Society expects from these gentlemen rare entertainment and instruction.

In our quotation from the *Nation's* Berlin letter on "The Discovery of the Soul," the printers made the mistake of not ending the paragraph with quotation marks, and our northern neighbors who copied it from the Journal entire without acknowledgement, have incorporated Sambo's philosophy in a way that we considered original with ourselves. It would be news to the *Nation's* correspondent that he is versed in the mysteries of the philosophy involved in the "folk lore" of our Southern negroes.

TO OUR READERS.

THE VALUE OF PURE WINE IN SICKNESS.

The chief difficulty with reliable wines has been their scarcity and exorbitant price, but this has been removed by the introduction of a Pure Native Wine, produced from the Oporto grape by Mr. Alfred Speer, of New Jersey. We understand he has submitted his Wine to the test of many of our celebrated physicians, and all concur in its purity, medicinal properties, and superiority to the best imported Port Wine. Most of them prescribe it in cases of debility, affections of the kidneys, and chronic complaints, requiring a tonic, sudorific or diuretic treatment.—*Examiner*. Salesroom 34 Warren street; N. Y.

MALTINE.

This above preparation is attracting the attention of the medical profession of Great Britain and the reputation it has acquired as one of our best nutritive agents is recognized and established, as the following extracts will show:

From the *British Medical Journal*, October 10th, 1878:—"At the late meeting of the British Medical Association, at Bath, in August last, among the visits of Pharmaceutical and Medical preparations, much interest was shown in one called *Maltine*, which may be described as a highly concentrated extract of *malted barley, wheat and oats*.

"Extracts of Malt (i. e., Malted barley) are pretty widely known, but this is the first example of a combination of the nutritious principles of these three cereals that we have seen, and the greater value of this combination is apparent, as wheat and oats are especially rich in muscular and fat-producing elements. This preparation is entirely free from the products of fermentation, such as alcohol and carbonic acid, and is very agreeable to the taste.

"Clinical experience enables us to recommend it as a nutritive and digestive agent, in virtue of its albuminoid contents, and its richness in phosphates and diastase, likely to prove an important remedy in pulmonary affections, debility, many forms of indigestion, imperfect nutrition, and deficient lactation. It will in many cases take the place of Cod Liver Oil and pancreatic emulsions, where these are not readily accepted by the stomach."

From the *Medical Times and Gazette*, November 2d, 1878:—"We have received a sample of a preparation called Maltine, which is described as being a concentrated extract of *malted barley, wheat and oats*. It is prepared with great care by a process fully described by the manufacturers (Reed & Carnrick) in a pamphlet which they will, we believe, willingly supply to any medical practitioner. The preparation possesses many qualities of great importance. It is non-alcoholic; it is agreeable to the taste; from its being so concentrated it is more portable than the liquid known as Malt Extract; and it possesses the virtues of that preparation in a much higher degree, inasmuch as it combines the principles of the three cereals above named,

and wheat and oats are especially rich in bone, fat, and muscle-producing elements. We have very good reasons for believing that it has been very carefully analyzed and examined by a competent authority, and proved to be very rich in diastase, in phosphates, and in albuminoid matters. It is very likely, therefore, to have considerable value as a digestive and nutritive agent in many wasting diseases, and cases of debility and imperfect assimilation. *Maltine* at any rate is well worthy of being put to the test practically by medical men, and it may be taken pure or mixed with water, wine or milk.”

UTERINE DISEASES.

By E. H. MURRELL, M. D., Lynchburg, Va.

It has been asserted that life is the most mysterious problem in nature, and that its duration is circumscribed by the laws of disease; consequently the incessant conflict which is waged between health and its insidious destroyer can never cease, but will ever continue to engage the attention of the chemist and pathologist in their profound researches of the best means of its preservation. Therefore, we need not be amazed or inquire why it is that the power of preserving life is held in such high estimation, or that the loss of it should be deemed a private misfortune or a public calamity.

Whatever may be the cause of constitutional debility and whenever prolonged, it must of necessity impair the healthy nutrition of the tissues, and lead to a low, inflammatory condition of the mucous membranes. The uterus and its appendages constitute no exception to the general rule, for whenever inflammation is lighted up in this organ, functional disturbance, accompanied by ulceration and hypertrophy follow as a natural consequence. On the other hand, it has been clearly demonstrated that constitutional debility of the gravest import and prolonged duration has its origin in those causes which induce sanguineous and muco-purulent discharges, continued pain, great despondency, and a general inability to partake of the accustomed food and exercise.

In the consideration of uterine diseases, including leucorrhœa, dysmenorrhœa, suppressed catamenia, menorrhagic and vaginal inflammation, it is proposed briefly to allude to the efficacy of the Bedford Alum and Iron Springs Mass as a curative agent, and to call attention to its tonic properties, which act most beneficially in their healthful restoration. By reference to the analysis of this Mass, it will be seen that it contains all the constituent properties most essential to the relief of morbid disease, namely by restraining the secretion while combining the tonic properties alike conducive to the improvement of the circulation and removing the causes which influence constitutional debility. For this reason, the water appears to exert a specific influence over the female organism, and often

displays its wonderful power of relief after the unsuccessful employment of all other remedial agents.

A brief synopsis of the treatment of suppressed catamenia by the Bedford Alum and Iron Springs Mass which came under the immediate attention of the writer, will suffice to attest its virtues:

Miss. J. S., a resident of this city, aged 20 years, of delicate constitution; had for months suffered from suppression of the catamenia which resulted in anæmia and great emaciation, attended with extreme nervousness, loss of appetite, constant cough, pain in the chest, night sweats, closely bordering on phthisis pulmonalis. After the exhaustion of numerous emmagogue agents which had been employed for months unsuccessfully, medical counsel was sought, and apparently with little hope of recovery. She was at once placed upon the free use of the Bedford Alum and Iron Springs Mass, which was continued for the space of three months, at the expiration of which time all organic disturbance was removed, with a complete subsidence of the symptoms before detailed, and a perfect restoration of her health, which has continued unimpaired to the present date.

Other cases might be adduced in testimony of the great efficacy of the Bedford Alum and Iron Springs Mass in the treatment of uterine diseases generally; but enjoying as it does so largely the public confidence and endorsed by the well-tested experience of the medical profession, any additional evidence in support of its virtues and wide-spread reputation would only prove superfluous and uncalled-for by the most skeptical.

OBITUARY.

ISAAC HAYS, M. D.

Dr. Isaac Hays, editor of the American Journal of Medical Sciences for 52 years, died in Philadelphia, Saturday, April 12th, 1879, 83 years of age.

WILLIAM A. DICK, M. D.

Dr. William A. Dick, formerly of Greensborough, N. C., and a son of the late Judge J. M. Dick, died in the town of Lumberton, N. C., on the 27th of March, 1879. Dr. Dick was no ordinary man. He was educated at the University of North Carolina, this institution conferring the degree of A. M. upon him in 1853. He graduated in Medicine at the University of New York, March, 1852; and began the practice of his profession in Lumberton in 1852. The medical profession has lost a bright light, and the community in which he lived a useful and honored citizen.

L.

GEORGE B. WOOD, M. D.

On the 30th ult., George B. Wood, M. D., of Philadelphia, died, in the 83d year of his age.

Dr. Wood was born at Greenwich, N. J., March 13th, 1797; graduated from the University of Pennsylvania as long ago as 1818. His extended career has been full of work, diligently pursued and successfully achieved. He was for many years Professor of Materia Medica in the Medical Department of the University of Pennsylvania, and from 1850 to 1860 was Professor of the Theory and Practice of Medicine in the same institution. At the time of his death he was President of the American Philosophical Society. He is the author of many important medical works, those by which he is best known being the "Dispensatory of the United States," written in collaboration with Franklin Bache, and first published in 1833, and his "Practice of Medicine," which long enjoyed great popularity. He also wrote much on historical subjects beyond the limits of his profession.

CARL F. BURKHARDT, M. D.

At a called meeting of the New Hanover County Medical Society to take suitable action in regard to the death of Dr. Carl F. Burkhardt.

The meeting was called to order by Dr. W. G. Thomas, President, and the following members were appointed to draft suitable resolutions: Drs. Love, Lane, and Wood.

It was resolved that the Society attend the funeral in a body.

WHEREAS, Dr. Carl Frederick Burkhardt has this day been stricken by the pitiless hand of death, we, the members of the New Hanover Medical Society, in honor of his memory and as an evidence of our esteem, adopt the following resolutions:

Resolved, That his professional attainments, his cultivated mind, his kind heart, his genial manners, deservedly won for him, who came among us a stranger, our respect and affection.

Resolved, That in his decease we lament a valued member of our profession, a good citizen and a gallant friend.

Resolved, That these resolutions be spread upon our minutes; that they be published in the NORTH CAROLINA MEDICAL JOURNAL, and in the daily press of this city; and that a copy duly attested, be presented to his family with assurance of our sympathy,

WM. J. LOVE,
WM. W. LANE,
THOMAS F. WOOD.

Wilmington, N. C., April 10th, 1878.

BOARD OF MEDICAL EXAMINERS.

To the Editors of the North Carolina Medical Journal:

In order to regulate the practice of Medicine and Surgery in the State of North Carolina, the General Assembly at the Sessions of 1858 and 1859, passed an Act to establish a Board of Medical Examiners.

Without a license from this Board, no physician who has commenced the practice of his profession in this State, “after the 15th of April, 1859, shall practice Medicine or Surgery, or any of the branches thereof, or in any case prescribe for the cure of disease, for fee or reward,” nor “shall he be entitled to sue for or recover, before any magistrate or court in this State, any medical bill for services rendered.” (See Laws of North Carolina, 1858–59.)

In conformity with the provisions of this Act, the State Board of Medical Examiners will meet in the city of Greensborough, on Monday, May 19th, 1879.

By authority of the Board.

PETER E. HINES, M. D., President.

HENRY T. BAHNSON, M. D., Secretary.

Salem, N. C., April 15th, 1879.

Medical Remuneration.—Doctor: “Um! most insolent!” (To his wife), “Listen to this my dear.” (reads letter aloud) “Sir, I inclose P. O. order for thirteen shillings and six pence, hoping it will do you as little good as your two small bottles of ‘physic’ did me.”—*Chicago Journal and Examiner.*—*Punch.*

BOOKS AND PERIODICALS RECEIVED.

Fern Etchings. By John Williamson. Specimen fasciculus.

A Clinical Treatise on Diseases of the Liver. By Dr. Freid. Theod. Frerichs. Vol. 2. Wm. Wood & Co.

Difficulties and Dangers of Battey's Operation. By George J. Engelmann, M. D., St. Louis, Mo. From Trans. Am. Med. Association, 1878.

Reports with Analyses of the Apollinaris Spring Neuenaha, Rhenish, Prussia. 1872. (Private and confidential).

Lectures on Practical Surgery. By H. H. Toland, M. D. Prof. of the Principles and Practice of Surgery. Second edition. Philadelphia. Lindsay & Blakiston, 25 South 6th Street.

Valedictory Address to the Graduating Class of Jefferson Medical College, at the Commencement March 12, 1879. By Prof. J. Aitken Meigs, M. D. 1879.

Ninth and Tenth Annual Reports of the Maryland Eye and Ear Institute. No. 66. N. Charles Street. Baltimore, Md. George Reuling, M. D. Surgeon in charge.

An Address upon the Life and Character of Lunsford Pitts Yandell, M. D. Delivered before the Kentucky Medical Society at the meeting held in Frankfort, April, 1878. By Richard O. Cowling, A. M., M. D.

Ringworm in Public Institutions. From Trans. Am. Medical Association. Rosacea. Extracted from the Transactions of the Medical Society of Pennsylvania. By John V. Shoemaker, M. D. Philadelphia, 1878.

The Causes of Sudden Death of Puerperal Women. An Address delivered before the American Medical Association, June 5th, 1878. By Edward W. Jenks, M. D. Chairman of Obstetrics and Diseases of Women and Children. Reprint from Trans. Am. Med. Association. 1878.

Maternal Impressions: Mothers Marks. An Exposé of a Popular Fallacy. By Roswell Park, A. M., M. D. Dem. of Anat. Chicago Medical College. Reprint from Southern Clinic. 1879.

A Manual of the Examination of the Eyes. A course of Lectures delivered at the "Ecole Pratique," by E. Landolt, Directeur, Adjoint, &c. Translated by Swan M. Burnett, M. D. Philadelphia. D. G. Brinton, 115 South Seventh Street. 1879. Price \$3.

[1.](#) Supposing the rain-fall to average thirty-six inches per annum, there is a fall of three cubic feet on every square foot of ground, or an acre receives twenty thousand nine hundred and eight (20,908) gallons of water per year.

[2.](#) In the river Loka in Sweden, and the Dead Sea, respectively; the latter containing four hundred thousand times more solid matter than the former. Ocean water has about 2,500 grains per gallon.

[3.](#) N. C. Medical Journal, Vol. 1. p. 177.

[4.](#) The American Journal of Obstetrics and Diseases of Women and Children, April 1879, p. 236-51.



TRANSCRIBER'S NOTES

1. Silently corrected typographical errors and variations in spelling.
2. Retained anachronistic, non-standard, and uncertain spellings as printed.
3. Footnotes have been re-indexed using numbers and collected together at the end of the last chapter.

*** END OF THE PROJECT GUTENBERG EBOOK NORTH
CAROLINA MEDICAL JOURNAL. VOL. 3. NO. 4. APRIL, 1879 ***

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